Akademia Górniczo-Hutnicza im. Stanisława Staszica w Krakowie

WYDZIAŁ INFORMATYKI, ELEKTRONIKI I TELEKOMUNIKACJI

Praca dyplomowa

Simulation Analysis of Multi-link Operation Performance in IEEE 802.11 Networks

Analiza symulacyjna wydajności mechanizmu Multi-Link Operation w sieciach standardu IEEE 802.11

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Kraków, 2026

I am sincerely grateful to my supervisor, coach and mentor, Professor Szymon Szott, for his great empathy, unbounded patience and continuous support, which helped me throughout the entire journey of writing this thesis.

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1. Introduction

Wireless local area networks (WLANs) based on the IEEE 802.11 (Wi-Fi) family of standards have become the dominant way of accessing the Internet indoors. Every new generation keeps pace with applications that are at once more bandwidth-hungry and more sensitive to delay, such as high resolution video streaming, cloud gaming, video conferencing as well as virtual and augmented reality (VR/AR), frequently in dense environments where many devices compete for the same resources. IEEE 802.11be, known as Wi-Fi 7, answers these demands with a set of enhancements at both the physical and medium access control layers [1, 2].

Among these enhancements, multi-link operation (MLO) is widely regarded as the most significant [3], allowing a single device to use several radio links at the same time, across the 2.4, 5, and 6 GHz bands, instead of being tied to a single channel. By aggregating the capacity of several links and being able to steer traffic away from a congested band, MLO promises higher throughput, lower latency, and better reliability. However, these benefits depend on the operating mode chosen for the device, in particular whether it can physically transmit on several links at once or whether it has to share a single radio. Other factors impacting MLO effectiveness include the offered load, the contention created by neighboring networks, and coexistence with existing single-link stations.

A substantial body of research has been undertaken to quantify what MLO actually delivers and what flaws are to be kept in mind. Most of these studies are based on analytical models, unspecified, implicit, or commercial simulators, or specific hardware; thus, the results are not always easy to compare. There is value in verifying them with an independent, openly available, and widely used packet-level simulator, both to check the reproducibility of the reported findings and to provide a single consistent framework in which different scenarios can be evaluated side by side.

Considering all of the above, the aim of the thesis is to analyze the performance of MLO in IEEE 802.11be networks by means of simulation using the ns-3 simulator, focusing on two of the MLO modes that the simulator implements, STR (simultaneous transmit and receive) and EMLSR (enhanced multi-link single radio), and comparing them against single-link operation. To achieve this goal, a representative selection of scenarios drawn from the recent literature is reproduced and extended, covering:

- the influence of the EMLSR configuration parameters on its performance;
- the maximum throughput achievable by the different transmission modes and how it changes with the channel width, frame aggregation settings, and the number of links;

- the latency, measured as the channel access delay, under varying load, number of stations, contention between networks, and channel occupancy;
- the ability of MLO to meet the strict delay requirements of virtual reality traffic;
- the coexistence of MLO devices with legacy single-link stations.

All scenarios are built on a common set of parameters and metrics so that their results remain directly comparable, and each one is related to the study it originates from, which allows the simulated behavior to be contrasted with the published one.

The remainder of this thesis is organized as follows. Chapter 2 provides the necessary background on Wi-Fi 7 and MLO, and reviews the state of the art. Chapter 3 describes the simulation model: the ns-3 simulator and its support for MLO, the traffic generators and simulation parameters, the performance metrics, and the data pipeline used to obtain the results. Chapter 4 presents and discusses the results of all the simulated scenarios, comparing them with the corresponding studies. Finally, Chapter 5 summarizes the findings and outlines possible directions for future work. Finally, all code developed as part of this thesis is available to the research community as open-source[1] to encourage reproducibility.

[1] *https://github.com/pawelwityk/mlo-wifi7-perf-tests*

2. Background

This chapter presents the background necessary for the rest of the work. The opening is based on a brief overview of Wi-Fi 7 and the directions in which it advances the IEEE 802.11 standard. The second section then turns to MLO, the mechanism whose performance this thesis measures, and describes its operating modes, its device architecture, and the ways it has been evaluated so far. The chapter closes with a review of the related work on which the simulated scenarios are based.

2.1. Wi-Fi 7

IEEE 802.11be, known as Wi-Fi 7, dubbed Extremely High Throughput (EHT), is one of the latest amendments to the IEEE 802.11 standard [1]. It pursues two main goals; a higher peak data rate, and a more predictable service, with lower latency and higher reliability. Both goals are driven by demanding applications such as high-resolution video streaming, cloud gaming, and virtual and augmented reality [2, 4]. To achieve these goals, the amendment extends both the physical (PHY) and the medium access control (MAC) layer, and its most relevant additions are described below, grouped by the goal they mainly serve.

2.1.1. Increasing Peak Data Rate

The largest gain in data rate comes from wider channels, as Wi-Fi 7 doubles the maximum channel width from 160 to 320 MHz. This change is accompanied by a denser modulation, 4096-QAM (4K-QAM), which adds two new modulation and coding schemes (MCSs), MCS 12 and MCS 13, and carries more bits in each symbol. Resource allocation becomes more flexible as well, since a single station can now be served by multiple resource units (when using orthogonal frequency-division multiple access, OFDMA), and preamble puncturing allows a wide channel width transmission to proceed even when part of the band is occupied, instead of falling back to a narrower channel. Together, these features raise the peak rate well above that of Wi-Fi 6, and they shape the throughput scenarios of this thesis, which rely on wide channels and high MCS values to measure the maximum performance of each transmission mode [2, 5].

2.1.2. Improving Latency and Reliability

The second group of features improves the timing of the service rather than its raw speed. The stream classification service (SCS) lets a station describe the requirements of its flows, and the restricted target wake time (rTWT) lets the access point (AP) reserve protected airtime for them, which reduces the jitter of latency-sensitive traffic in managed networks. The most far-reaching addition, however, is MLO, together with the traffic-identifier-to-link mapping (T2LM) that selects which link carries which flow, for instance reserving the 6 GHz band for delay-critical traffic [4]. Since MLO is the feature studied in this work, it is treated in detail throughout the rest of the chapter.

2.2. Multi-Link Operation

MLO lets a single logical Wi-Fi device use several links at once, across the 2.4, 5, and 6 GHz bands, where every earlier Wi-Fi generation was confined to one band at a time. A device that supports it, whether an AP or a station, is called a multi-link device (MLD), in contrast to the legacy single-link devices (SLDs) that still make up most of the installed base. The mechanism is expected to help for two reasons: several links together offer more capacity than one, and a device that can choose between links can move its traffic away from a channel that has become congested. The gain a given MLD obtains, however, depends on how it is permitted to use its links and, in particular, on whether it can be active on all of them at once or only one at a time. The standard captures this through a set of operating modes.

2.2.1. Operating Modes

IEEE 802.11be groups the MLO modes according to the number of radios the device has, as shown in Figure 2.1. A multi-radio device carries one full radio per link and can therefore be active on several links at once, whereas a single-radio device has only one fully capable radio and must share it between links.

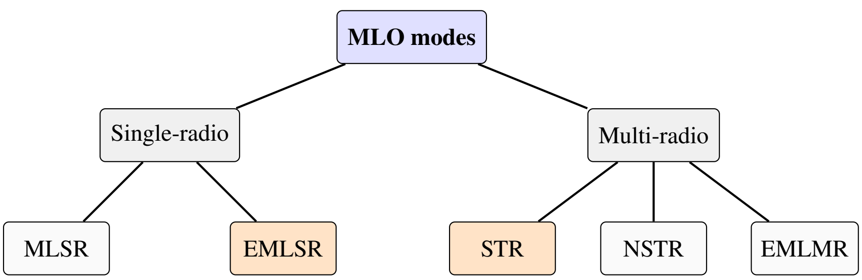


Fig. 2.1. Categorization of the MLO modes by the number of radios [1]. The two modes highlighted in orange (EMLSR and STR) are the ones studied in this work.

The single-radio category contains two modes that differ in how the shared radio is managed:

– *Multi-Link Single Radio* (MLSR) keeps its single radio running on one link and switches it to another only when needed. Because the radio listens on one link at a time, the AP must know which link is currently active before it can reach the station, and every switch adds a switching delay.

– *Enhanced Multi-Link Single Radio* (EMLSR) removes this restriction by pairing the main radio with several low-capability auxiliary radios that keep listening on the other links, just enough to sense the channel and decode control frames. When the AP starts a transmission on one of these links with an initial control frame, the main radio is brought over to handle the data. The time it needs to switch is announced to the AP as a padding delay added to the control frame [6, 7]. EMLSR is the single-radio mode available in the simulator and used in this work.

The two modes differ in reachability: MLSR ties its single radio to one link, whereas EMLSR keeps every link under observation and brings the main radio over on demand, paying a short switching time in exchange for being able to use whichever link becomes free first. The EMLSR frame exchange is shown in Figure 2.3.

Meanwhile, for more capable devices, the multi-radio category provides three modes:

– *Simultaneous Transmit and Receive* (STR) gives each link its own radio and its own independent backoff, so that a transmission on one link and a reception on another can proceed at the same time. This independence is what makes STR multiply the throughput by the number of links and keep the channel access delay low. Its main drawback is in-device coexistence interference, i.e., the power that leaks between links that are not separated far enough in frequency. STR is the multi-radio mode available in the simulator and used in this work.

– *Non-Simultaneous Transmit and Receive* (NSTR) makes one link primary and the others dependent on its state, and does not permit simultaneous transmission and reception on different links. Giving up the simultaneity removes much of the in-device interference that affects STR, at the cost of some flexibility.

– *Enhanced Multi-Link Multi Radio* (EMLMR) goes a step further and allows the device to move its radio resources, such as its spatial streams, from one link to another as demand changes.

Figures 2.2 and 2.3 illustrate the difference between the two modes that allow concurrent operation. In STR, the two links work independently, so the device may transmit on one while receiving on the other, in opposite directions. In NSTR, the transmissions on the two links are instead aligned in time and direction, which is what keeps the device from transmitting on one link while receiving on the other, as shown in Figure 2.4.

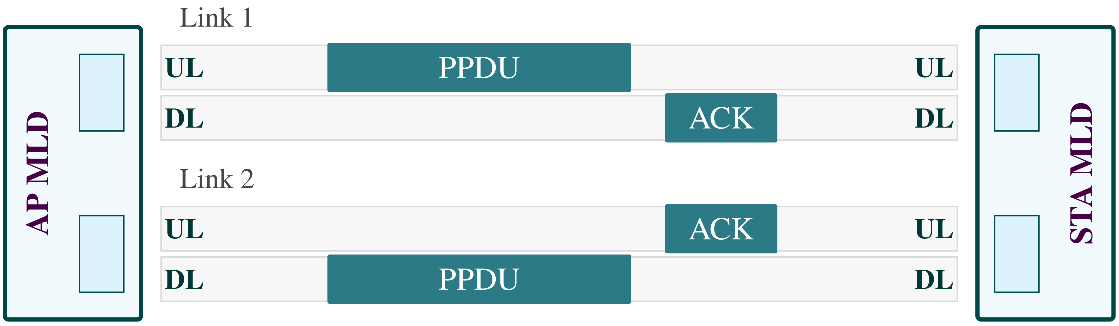


Fig. 2.2. Example frame exchange in STR mode. Each link has its own radio, so the device transmits a PPDU on link 1 (uplink) while at the same time receiving a PPDU on link 2 (downlink); each link returns its own acknowledgement in the opposite direction.

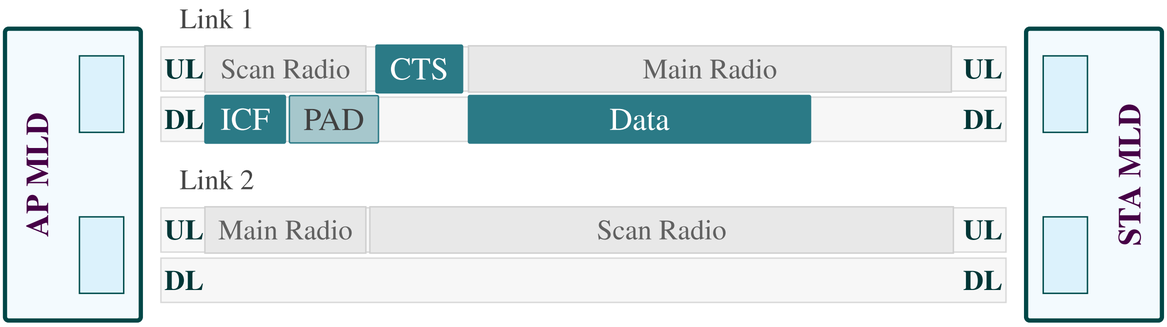


Fig. 2.3. Example frame exchange in EMLSR mode. The auxiliary (scan) radios listen on both links. When the AP sends the initial control frame (ICF) and a padding interval (PAD) on link 1, the station moves its main radio there (leaving link 2 covered only by a scan radio), answers with a CTS, and the data exchange then proceeds on link 1.

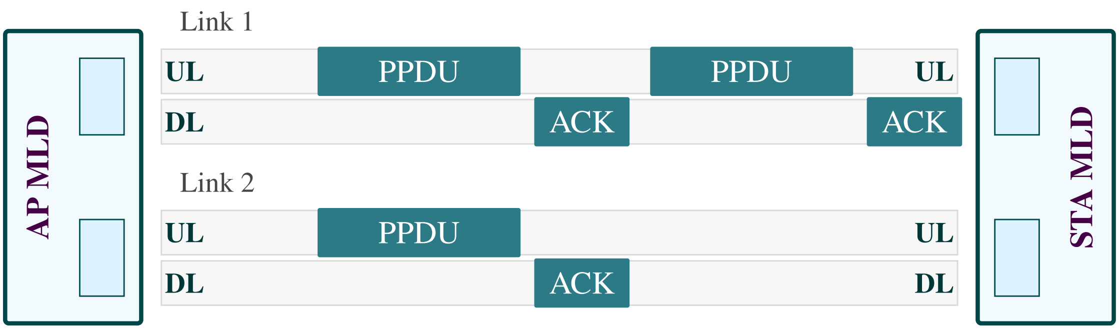


Fig. 2.4. Example frame exchange in NSTR mode. The transmissions on the two links are aligned in time and share the same direction (uplink data, downlink acknowledgement), so the device never transmits on one link while receiving on the other.

2.2.2. MLO Device Architecture

The presentation of several radios to the upper layers as a single device requires a modified MAC [3, 8]. Therefore, Wi-Fi 7 divides this MAC into sublayers. The upper sublayer is common to all links and performs link-independent functions, such as frame aggregation, sequence numbering, encryption, and mapping of traffic identifiers to links. The lower sublayer is repeated for each link and handles the channel access on that link. What the standard does not define is how frames should be scheduled across the links from one moment to the next; it leaves this policy to implementation, and as the related work shows, this open choice has become a research topic for several studies [9, 10, 11, 12, 13, 14, 15].

2.2.3. Approaches to Evaluating MLO

Because MLO touches every layer from the physical to the application, it has been studied with a range of complementary methods, each striking a different balance between simplicity and realism. Analytical and Markov-chain models compute the saturated throughput and the mean access delay of a few links with little effort, but they rest on simplifying assumptions, such as saturation and ideal channel, and independent links, which limit how far their conclusions carry to realistic traffic and contention. Measurements on real Wi-Fi 7 hardware are the most faithful, but they are tied to a particular device type, model, revision, etc., and are hard to reproduce. In addition, the potential cost of these measurements could be unacceptable. Between those two extremes lie system-level simulators, which model the MAC and an abstracted PHY in enough detail to run complete scenarios while remaining repeatable and cost-effective.

Two simulators dominate the literature. The MATLAB WLAN Toolbox offers a detailed, vendor-maintained 802.11be model and is the basis of a recent master thesis which this work parallels [16]. However, it is a closed commercial environment and simulation runs are considerably long. Meanwhile, in the studies performed by Carrascosa-Zamacois et al. [17, 18], an undisclosed simulator is used, which was verified by comparing the results with the calculations from an analytical Markov model. Without a doubt, the biggest disadvantage is the lack of ability to reproduce the simulations, due to the restricted access to the tool used. The ns-3 network simulator, on the contrary, is open-source, free and widely used in networking research, and its 802.11be module has matured to include a complete multi-link MAC together with STR and EMLSR modes [7]. This thesis adopts ns-3 for several reasons: being open, it lets every scenario be reproduced and shared, and a single simulator can host a whole range of experiments, from throughput and latency to coexistence and application traffic, under one consistent set of assumptions. Reproducing in ns-3 results first obtained analytically, in MATLAB, or in hardware is valuable in itself, since agreement between independent tools strengthens confidence in a finding, while disagreement points to a modeling assumption worth a closer look. The latter is exactly what happens for one of the latency scenarios presented later in this work.

2.3. State of the Art

Wi-Fi 7, and MLO in particular, have prompted a large body of research, most of it concerned with how much MLO improves the quality of service over SLO. A recurring question is the size of the throughput and latency gains. Jenkić and Kočan [19] study downlink performance in a network with 1, 4, or 10 stations and report that two links improve throughput by about 92% on 80 MHz channels and 84% on 160 MHz channels relative to a single link, while also lowering the average packet latency; they further note that two 80 MHz links outperform a single 160 MHz channel. Broader evaluations of the standard [5] and of the individual MLO modes under varying overlapping-BSS activity [17] confirm that all MLO modes increase the throughput over SLO, with STR benefiting the most, since it alone can use several links in parallel.

Another line of work looks at the problems that MLO can create in dense, contended deployments. Carrascosa-Zamacois et al. [18] identify a latency anomaly, related to a starvation effect, in which an STR device that occupies all its links blocks neighboring BSSs and can end up with higher maximum latency than an SLD. They show that the problem is best contained by careful channel assignment, for example, by giving each BSS one private link in addition to a shared one, or by using more links than there are contending BSSs. Driving the overlapping traffic with real 5 GHz spectrum-occupancy traces, the same group [20] finds that MLO still outperforms SLO in most conditions, but that two asymmetrically occupied links can degrade performance when a packet is assigned to a link before the backoff has been resolved; to avoid this, they propose an alternative channel access scheme that runs parallel backoffs. The coexistence of MLO with legacy SLDs has since been studied more formally: Matsuda et al. [21] extend the classical Markov-chain throughput analysis to networks that mix MLO and SLO stations, and Gao et al. [22] derive an analytical model of the end-to-end delay, jitter, and maximum latency of such coexisting networks, together with the traffic-to-link allocation that minimizes them, showing through ns-3 simulation that earlier allocation policies become noticeably worse once legacy SLDs are present. Such analytical treatments are part of a broader effort to model the behavior of MLO in closed form, including Markov-chain models of its throughput and service quality on Wi-Fi 7 [23].

A further, active research direction concerns the policy that decides how traffic is distributed across the links, which the standard leaves unspecified. López-Raventós and Bellalta show that, depending on the load, the best choice may even be to use a single, least-congested interface [9], and later propose congestion-aware policies that periodically rebalance the load across links [10]. Building on this idea, later works tailor the allocation to particular goals: legacy-friendly strategies that protect coexisting SLO stations [11], and service-level-aware strategies that bound the per-flow-latency in industrial or low latency settings [12, 13]. More recent studies extend the objective beyond latency alone. Shen et al. [24] study the latency-power trade-off of different allocation mechanisms, and the related question of which AP a station should join when links have unequal capabilities has been addressed with a centralized, network-controlled allocation scheme [25]. Scheduling is refined in the same spirit: channel occupancy information can drive an MLO scheduler in ns-3 [14], while the restricted target wake time introduced in Wi-Fi 7 can be coordinated across links to deliver predictable, low-jitter latency [15]. Closely related efforts tune the EDCA parameters and the assignment of access categories to links to meet delay-bounded QoS targets [26], and study how packet steering can be realized in the packet-processing pipeline of real Wi-Fi chipsets [27]. Across this line of work the same message recurs: the choice of policy can matter as much as the decision to use MLO at all, and that choice is left to the implementation.

A complementary strand of research treats MLO as a means of reliability, rather than raw performance. Because a multi-link device can send a frame on several links at once, it can carry out frame replication and elimination, which Haxhibeqiri et al. [28] use to extend the guarantees of time-sensitive wired networking to a wireless segment. Reliability, however, interacts with the reordering that several parallel links introduce: Oliveira et al. [29] show that out-of-order delivery across links can stall the block-acknowledgment window and create latency spikes at the application layer, an effect that is especially harmful to TCP. The question of coexistence, finally, extends beyond Wi-Fi itself, as the behavior of NSTR MLDs that share unlicensed spectrum with cellular LTE-LAA has been implemented and evaluated in ns-3 [30].

Other studies target specific mechanisms and use cases. On the physical side, Paroshin et al. [31] observe that distributing MPDUs unevenly across links starves the underutilized ones, and propose a dynamic A-MPDU sizing algorithm that raises the throughput by up to 50%. On the coexistence side, Medda et al. [32] examine how MLO devices share the medium with the large installed base of legacy SLO stations, a concern about inclusion and backward compatibility that runs through the transition to Wi-Fi 7. At the transport layer, the interaction between MLO and TCP, whose reliance on acknowledgments is sensitive to how links are used, has been studied both for throughput friendliness [33] and for latency-sensitive applications [34]. The suitability of MLO has also been assessed for a range of applications: against the strict delay limits of extended-reality streaming [35] and augmented-reality workloads [36], for the quality of experience of video and audio streaming steered through traffic-to-link mapping [37], and for the dense Internet of Things deployments that Wi-Fi 7 is expected to serve [38]. Looking further ahead, MLO continues to evolve towards the next Wi-Fi generations: it is being combined with non-orthogonal multiple access to increase the throughput of the forthcoming Wi-Fi 9 (IEEE 802.11bn) [39], and learning-based methods, such as federated, transformer-based traffic-to-link mapping, are being explored for 802.11bn Internet of Things networks [40].

It should be noted that a growing share of this recent work relies on the same open simulators discussed earlier in this chapter, with ns-3 used in several of the coexistence and scheduling studies [14, 30, 22] and OMNeT++ in others [15], which makes the comparison between tools pursued here more accurate. The scenarios reproduced in this work are drawn from this body of research and cover its throughput, latency, coexistence, and applications strands. They are evaluated within a single ns-3 framework, so that published findings can be set against the output of an independent simulator.

3. Simulation Model and Setup

This chapter presents the simulation model used throughout the study. Firstly, the ns-3 simulator and its support for MLO are described. Then, we describe the traffic generators and parameters common to all scenarios, the metrics used to quantify performance, and finally the data pipeline that turns the raw simulation output into the figures presented in the next chapter.

3.1. Network Simulator

The ns-3 simulator is used because it provides a variety of tools that can be used to simulate MLO characteristics, enabling researchers to verify its advantages and disadvantages compared to basic SLO scenarios. The following subsections contain information about ns-3, its general capabilities, and support for Wi-Fi and MLO.

3.1.1. Overview of ns-3

The ns-3 simulator is an open-source computer network simulator developed by a community of users as a successor to the older ns-2 simulator. Since ns-3 is a discrete event simulator (DES), it can accurately reflect the characteristics of packet-switched networks. The primary target audience of ns-3 consists of students and researchers who use it for scientific and educational purposes. The simulator supports network technologies such as Ethernet, LTE, 5G, Wi-Fi, but also includes mobility models, traffic models, propagation models, etc.

Unlike its predecessor, ns-3 is written entirely in the C++ programming language. Simulation scenarios can either be written directly in C++ or using Python wrappers. The modular architecture of the simulator allows its functionality to be easily extended by independent developers. The goal of ns-3 developers is to provide clear documentation and faithful representation of real-world data transmission characteristics in wired and wireless networks, making the simulator widely used in educational activities, scientific research, and simulations of complex networking environments. In summary, the main advantages of ns-3 include the following:

- availability of the source code,
- an actively developed user community,

– regular updates,

– broad support for various networking technologies,

– the ability to accurately reproduce the operation of network protocols,

– relatively simple configuration of simulation scenarios.

3.1.2. Wi-Fi in ns-3

The Wi-Fi module in ns-3 is the most complex and developed part of the entire simulator's environment, providing support for a variety of 802.11 standards to simulate radio transmission and analyze its performance under different propagation conditions. The simulator's current version (ns-3.47) supports the whole IEEE 802.11 family up to and including the 802.11be (Wi-Fi 7) amendment, that is, the legacy 802.11a/b/g standards together with the later 802.11n, 802.11ac, and 802.11ax generations. The PHY/MAC standard for a device is selected with a single call shown in Listing 3.1.

```
wifi.SetStandard(WIFI_STANDARD_80211be);
```

Listing 3.1. Selecting the IEEE 802.11 standard in ns-3.

Within the configuration process, the user can define many parameters related both to the physical layer and to the MAC layer. The most relevant ones for this work are described in the following, illustrated with the code actually used in the simulations.

Channel number and width. Each link is given an operating channel through the `ChannelSettings` attribute, whose value is a tuple of the channel number, the channel width in MHz, the frequency band, and the index of the primary 20 MHz channel. As shown in Listing 3.2, link 0 operates on channel 42, with an 80 MHz width, in the 5 GHz band. Changing the second field (for instance, to 160) widens the channel, while the band field places the link in the 2.4, 5, or 6 GHz spectrum, which is exactly how the different bands of an MLO device are assigned.

```
phy.Set(0, "ChannelSettings", StringValue("{42, 80, BAND_5GHZ, 0}"));
```

Listing 3.2. Configuring channel settings for a Wi-Fi link in ns-3.

Transmission power. The transmit power, expressed in dBm, is configured per link through the `TxPowerStart` and `TxPowerEnd` attributes. Setting both to the same value, as shown in Listing 3.3, fixes the power at 20 dBm; a non-empty range between the two would expose several discrete power levels to the transmit power control mechanism.

```
phy.Set(linkId, "TxPowerStart", DoubleValue(20.0));
phy.Set(linkId, "TxPowerEnd",   DoubleValue(20.0));
```

Listing 3.3. Configuring the transmission power.

Modulation and coding. The MCS is chosen through the rate control manager, as shown in the Listing 3.4. All reported simulations use `ConstantRateWifiManager`, which keeps the MCS fixed so that the results are not blurred by rate adaptation. Data frames are sent with high-order `EhtMCS11`, whereas control frames use `EhtMcs0`, which is representative of real-world settings.

```
wifi.SetRemoteStationManager("ns3::ConstantRateWifiManager",
                             "DataMode",    StringValue("EhtMcs11"),
                             "ControlMode", StringValue("EhtMcs0"));
```

Listing 3.4. Setting up the data and control plane MCS values.

Spatial streams. The MIMO configuration is set per link by the number of antennas and the supported number of spatial streams, as shown in the Listing 3.5. This configuration defines a 2×2 MIMO link, which roughly doubles the physical-layer rate compared to a single stream.

```
phy.Set(i, "Antennas",                      UintegerValue(2));
phy.Set(i, "MaxSupportedTxSpatialStreams", UintegerValue(2));
phy.Set(i, "MaxSupportedRxSpatialStreams", UintegerValue(2));
```

Listing 3.5. Configuring the antennas and spatial streams supported.

Frame aggregation. Wi-Fi's A-MPDU aggregation is controlled at the MAC through the maximum A-MPDU size and the MPDU buffer size. In Listing 3.6, up to `nMpdus` best effort frames are aggregated into a single A-MPDU; increasing this limit (for example, from 64 to 1024) increases the aggregation and, with it, the achievable throughput, as examined in the throughput scenarios in Section 4.2.

```
mac.SetType("ns3::ApWifiMac",
            "BE_MaxAmpduSize", UintegerValue(nMpdus * (payloadSize + 60)),
            "MpduBufferSize",  UintegerValue(nMpdus));
```

Listing 3.6. Choosing the A-MPDU aggregation mode.

Path loss. An important element of wireless network simulations is the propagation model responsible for reproducing radio signal attenuation. Among the available path loss models are:

– `FriisPropagationLossModel` – this model is based on the Friis free-space propagation equation and represents ideal signal propagation conditions without obstacles, reflections, or interference. Signal attenuation depends only on the distance between the transmitter and receiver, as well as on the carrier frequency. The model is mainly used for open-space environments and theoretical analysis.

– `LogDistancePropagationLossModel` – the Log-Distance model extends the free-space propagation model by introducing a path loss exponent that describes how quickly the signal attenuates with distance. The model is commonly used for realistic wireless environments, such as urban, suburban, and indoor scenarios where obstacles and reflections affect signal propagation.

– `NakagamiPropagationLossModel` – this model is used to simulate multipath fading effects in wireless communication channels, by statistically modeling rapid signal fluctuations caused by reflections, scattering, and diffraction. The Nakagami model is widely used in simulations of mobile and indoor wireless networks due to its flexibility in representing different fading conditions.

– `ThreeLogDistancePropagationLossModel` – this model is an extension of the Log-Distance model and divides the propagation environment into three distance regions, each with a different path loss exponent. This allows for more accurate modeling of signal attenuation over short, medium, and long distances in complex environments.

These models make it possible to analyze the impact of distance, obstacles, and multipath fading on transmission quality. The simulations in this work use the log-distance model, as it is the most suitable for residential conditions, and broadly used in the variety of studies. The model is instantiated and attached to the channel as shown in Listing 3.7. An exponent of 3.5 (against 2.0 for free space) together with a 40 dB loss at the 1 m reference distance reproduces a residential indoor environment in which walls and furniture attenuate the signal faster than in open space; increasing the exponent corresponds to a more obstructed environment.

```
Ptr<LogDistancePropagationLossModel> loss =
    CreateObject<LogDistancePropagationLossModel>();
loss->SetAttribute("Exponent",          DoubleValue(3.5));
loss->SetAttribute("ReferenceLoss",     DoubleValue(40.0));
loss->SetAttribute("ReferenceDistance", DoubleValue(1.0));
spectrumChannel->AddPropagationLossModel(loss);
```

Listing 3.7. Configuring a pathloss model.

Channel sharing. Interference between networks is implicitly reproduced through the shared transmission medium. Two BSSs operating on the same physical channel are attached to the same spectrum-channel object, as presented in Listing 3.8, so that each one receives the transmissions of the other, and the two must contend for the medium. By mapping several BSSs onto the same `specMedia[]` object, the simulation reproduces the contention and co-channel interference that arise when neighboring networks share a channel – precisely the mechanism behind the coexistence scenarios (Section 4.5). Such scenarios make it possible to analyze the performance of modern Wi-Fi standards in conditions close to real-world deployments.

```
phy.AddChannel(specMedia[], phyChans[].range);
phy.Set(l, "ChannelSettings", StringValue(chSet.str()));
```

Listing 3.8. Setting up the BSS shared medium.

3.1.3. MLO Support

Along with the development of the Wi-Fi 7 module in ns-3, support for the MLO mechanism, which is one of the most important features of the IEEE 802.11be standard, has gradually been introduced into the simulator. The first introduction was performed in version 3.38 with just the STR mode. This addition of the first MLO mode allowed initial research on the operation of devices using multiple radio links simultaneously.

The current MLO implementation in ns-3.47 enables simulation of data transmission using multiple frequency bands at the same time, including 2.4, 5, and 6 GHz, for analyzing the impact of multi-link transmission on throughput, latency, and reliability of wireless communication. The simulator allows configuration of two operating modes for multi-link devices: STR and EMLSR. Depending on the selected mode, it is possible to study the behavior of devices equipped with multiple independent radio chains (STR) or a single shared radio interface that switches between channels (EMLSR). The ns-3 simulator does not implement a dedicated multi-link scheduler; instead, each link runs its own independent EDCA backoff, so frames are transmitted on whichever link wins channel access first. Support for MLO in ns-3 is implemented by extending both the PHY and MAC layer models. For an MLO device the user can define, among others:

- the number of active links (the link count passed to `SpectrumWifiPhyHelper`);
- the channel of each link, i.e., its number, width, and frequency band (`ChannelSettings`);
- the operating mode of the device, STR or EMLSR (`EmlsrActivated` together with `SetEmlsrManager`);
- the EMLSR parameters, such as the set of EMLSR links and the behaviour of the auxiliary radios (the attributes of the EMLSR manager, e.g. `EmlsrLinkSet`, evaluated later in the EMLSR parameter testing scenario);
- the mapping of traffic flows to links, through the TID-to-link mapping (`SetTidLinkMapping`).

These options make it possible to study how the MLO performance of a Wi-Fi 7 network is affected by:

- the co-channel interference between coexisting networks,
- the offered network load,
- the number of active stations,
- the channel occupancy of the individual links,
- the operating mode and the way the links are assigned to the BSSs.

3.2. Traffic Generators and Simulation Parameters

Every scenario is assembled from the same set of components: a number of traffic generators and a configuration that is, for the most part, shared across all simulations. Only the values that genuinely distinguish a scenario – such as the number of links, the channel bandwidth, or the operating mode – are changed and reported in the corresponding scenario description; everything else is kept fixed so that the results remain comparable.

3.2.1. Traffic Generators

Two families of traffic generators are used, depending on whether a scenario targets throughput or latency:

- *Constant-rate UDP* (`UdpClientHelper` with `UdpServerHelper`) is used in Sections 4.1 and 4.2. The client generates fixed-size datagrams with a fixed inter-packet interval time. For the maximum-capacity measurements, this interval is set to a single microsecond, which corresponds to an offered load of roughly 12 Gbit/s (for a payload of 1500 bytes) and therefore drives all links into saturation. For the normalized-load scenarios, the interval is calculated to match the desired offered rate.
- *Constant On-Off* (`OnOffHelper` with an effectively infinite on-time and a zero off-time) is used in the latency and coexistence scenarios. The generator produces a steady stream at a configured `DataRate`, which reproduces the constant-bit-rate traffic of the referenced works.
- *VR On-Off* drives the virtual-reality scenario with a bursty pattern: an 11.1 ms period (about 90 frames per second) with a 50% duty cycle [35], so that the peak rate is twice the average. Per-packet timestamps are enabled (`EnableSeqTsSizeHeader`) so that the end-to-end delay of every frame can be recovered at the receiver.
- Background traffic for the overlapping BSSs and for the legacy-coexistence stations is generated by additional on-off sources, which load the competing devices to a controlled level.

Rather than absolute data rates, most scenarios express the offered load as a normalized value $\rho \in [0, 1]$, defined as a fraction of the SLO saturation throughput measured in Section 4.2.1 (for example, approximately 1150 Mbit/s on an 80 MHz channel and approximately 2200 Mbit/s on a 160 MHz channel for A-MPDU = 1024 MPDUs). A value of $\rho = 1$ therefore means that the offered load equals that reference peak, which makes the load directly comparable across configurations with different physical layer rates.

3.2.2. Common Simulation Parameters

The fixed part of the configuration follows the residential setup used by the reference studies. The physical layer is the spectrum-based abstraction (`SpectrumWifiPhyHelper` over a multi-model spectrum channel), configured for the IEEE 802.11be (EHT) standard with two antennas (so two spatial streams) and an 800 ns guard interval. The rate is fixed with `ConstantRateWifiManager` at a chosen EHT MCS, with control frames sent at the most reliable modulation. Frame aggregation uses A-MPDU with a configurable buffer size (64, 512 or 1024 MPDUs). Signal attenuation is modeled with the `LogDistancePropagationLossModel` (path-loss exponent 3.5 and 40 dB reference loss at 1 m). The AP is placed with stations on a circle around it, at a distance of a few meters, using a constant-position mobility model. Each data point is averaged over ten independent replications obtained with different `RngRun` seeds. These shared values are later repeated, for convenience, in Table 4.3, while the scenario-specific deviations are listed alongside each scenario.

3.3. Performance Metrics

Two primary metrics are used to characterize performance, throughput and latency. Both are computed from traces collected during the simulation rather than from a single aggregate counter, allowing both mean values and full distributions to be reported.

3.3.1. Throughput

Throughput is measured at the MAC layer as the total amount of payload successfully delivered to the receiving station, recorded through the `MacRx` trace and divided by the measurement window:

$$\text{Throughput} = \frac{8 \sum_i b_i}{T \cdot 10^6} \quad [\text{Mbit/s}], \tag{3.1}$$

where b_i is the size in bytes of the i-th frame delivered to the station's MAC and T is the duration of the measurement. As a cross-check, the same quantity is also obtained at the network layer from `FlowMonitor`, which accounts the received bytes of each flow over its active lifetime.

3.3.2. Latency

The latency metric is the *channel access delay*, defined as the time elapsed between the moment when a packet is passed to the MAC layer and the moment when its transmission actually starts at the physical layer. For each packet k, this delay is computed as

$$d_k = t_k^{\text{PhyTxBegin}} - t_k^{\text{MacTx}}, \tag{3.2}$$

where t_k^{MacTx} is the timestamp of the `MacTx` trace and $t_k^{\text{PhyTxBegin}}$ is the timestamp of the corresponding `PhyTxBegin` trace for the same packet. The two timestamps are matched using the packet identifier.

Thus, d_k represents the time spent by packet k inside the MAC layer before it gains access to the wireless medium. This interval captures the effects of contention, random backoff, and, in the EMLSR case, link-switching time. These are precisely the components that MLO mechanisms are expected to influence. The delay is reported as the mean value as well as the 50th and 99th percentiles, computed from the sorted set of per-packet delay samples.

3.4. Data Pipeline

Each scenario is organized as a self-contained unit consisting of an ns-3 simulation written in C++, a Python runner that drives it, and one or more plotting scripts, all relying on a small set of shared helpers. The flow from the simulation code to the final figures is summarized in Figure 3.1.

The C++ code implements a single configuration: it takes the variable parameters as command-line arguments, runs one simulation, and prints the resulting metrics as labeled lines in its standard output. The Python runner enumerates the full parameter list, multiplies it by the ten random seeds, and launches the optimized binary as independent subprocesses on a pool of worker threads, so that many configurations are simulated in parallel. Each metric is parsed from the simulation output with regular expressions and a result file is stored for each configuration and seed. Because the results are stored individually, an interrupted run can be resumed without recomputing the finished points, and any job that fails is retried automatically.

Finally, the plotting and aggregation scripts read the stored result files, average each metric over the random seeds, attach a 95% confidence interval (computed with the Student-t distribution given the small number of replications), and render every figure through a shared style module so that all the figures in this work are visually consistent. Their individual styling can differ because the main goal is to match them with their equivalents in studies on which the scenarios are based.

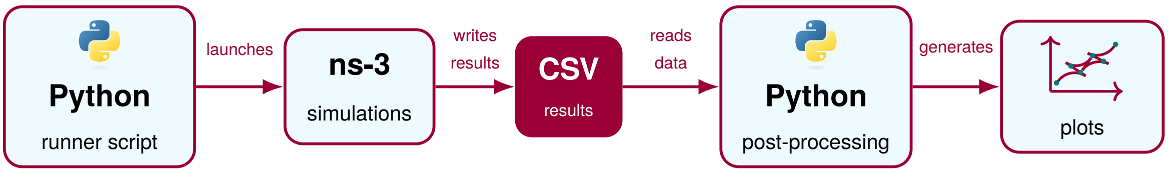


Fig. 3.1. Workflow used in the simulation environment.

4. Simulation Results

This chapter presents results from multiple simulation scenarios which examine MLO performance under various conditions. The results of these simulations mainly investigate MLO's potential to increase throughput or reduce latency. Additional scenarios address VR performance, coexistence of MLO with legacy stations, and an evaluation of EMLSR's configuration parameters. All analyzed scenarios are based on recent studies in the MLO field. The chapter concludes with a discussion of the achieved results.

4.1. Preliminary EMLSR Study

Because the implementation of EMLSR in ns-3 has a variety of parameters that can be modified, it is crucial to measure how these parameters affect its performance, so that the rest of the EMLSR experiments in this work can be run with justified settings. The measurements are carried out in a single BSS with one AP and either one or four stations, connected over two EMLSR links, one in the 5 GHz band and another in the 6 GHz band, both 80 MHz wide. The AP sends constant rate downlink traffic at a fixed normalized load of 0.7, in which the system is not yet saturated. The fixed part of the configuration is given in Table 4.1, while the swept parameters, their tested ranges, and the impact that each turned out to have are collected in Table 4.2.

Table 4.1. Fixed simulation parameters for the EMLSR parameter evaluation.

Simulation parameter	Value
A-MPDU size	1024 MPDUs
Band	5 & 6 GHz
Channel width	80 MHz
Channels	5 GHz: 42, 6 GHz: 7
MCS	11
Normalized load	0.7
Number of links	2
Number of stations	1, 4

The study separates the parameters into three groups according to their impact (Table 4.2). The settings in the first group (transition timeout, auxiliary PHY width, and auxiliary sleep flag) have no measurable effect on this test, as the latency values remain unchanged in spite of values set, and can be chosen freely. The settings in the second group (transition delay and padding delay) do not have an impact on throughput but add an access delay in proportion to the parameters' value, as expected from parameters that model switching overhead. The padding delay is the more costly of the two, raising the access delay from about 0.7 to 1.3 ms at its highest value. Settings belonging to the third group (auxiliary TX capability and auxiliary PHY switching flag) have the greatest impact on performance. Allowing the auxiliary radio to transmit lowers the access delay, most visibly with a single station. Letting the auxiliary PHY follow the main radio on a link switch is by far the most influential parameter of the whole study: with four stations it raises the throughput by roughly 14% and cuts the access delay by about 80%, since leaving it off strands one link without a listening radio after every switch and breaks the symmetry on which EMLSR relies.

Table 4.2. Impact of EMLSR parameters on downlink throughput and channel access delay.

Parameter	Tested values	Preferred	Impact
Transition delay	0–256 μs	0 μs	Throughput unaffected; access delay grows slowly with the value.
Padding delay	0–256 μs	0 μs	Throughput unaffected; adds the most access delay of the two delay parameters (about 0.7 to 1.3 ms).
Transition timeout	128–16384 μs	any	No measurable effect on either metric.
Aux PHY width	20–80 MHz	80 MHz	No measurable effect in this test.
Aux PHY sleep	off / on	off	No measurable effect; off keeps the aux alert, on saves power.
Aux PHY TX-capable	off / on	on	Throughput changes indistinguishable from noise; lower access delay when on, mainly with one station.
Switch aux PHY	off / on	on	Most influential: about 14% higher throughput and 80% lower access delay with four stations.

These results justify the EMLSR baseline adopted in the rest of this work: the transition delay and padding delays are set to zero, while the auxiliary PHY is kept awake, made TX-capable, and allowed to follow the main radio; the transition timeout and the auxiliary width are left at their default values due to their lack of impact on the results.

4.2. Throughput

The first main set of scenarios measure the throughput performance of MLO. This metric gives an opportunity to clearly show the advantages of MLO over SLO transmission with regard to the ability to use multiple links simultaneously. The scenarios open with a measure of the maximum capabilities of different transmission modes that are later used as a reference point to which the rest of the tested cases can be compared. Then a variety of tests show how different MLO settings behave in a number of conditions. All of these scenarios are based on the same basic settings (Table 4.3). Any changes to these settings are clearly stated in the scenario descriptions.

Table 4.3. Basic simulation parameters.

Simulation parameter	**Value**
PHY abstraction class	`SpectrumWifiPhyHelper`
Spatial streams	2
Simulation time	10 s
Number of independent runs	10
Pathloss model	Log-distance
A-MPDU size	1024 MPDUs

4.2.1. Throughput Scenario 1

The first scenario, based on [19], provides the reference point for all other scenarios, giving the maximum capacities of all of the used transmission modes, and configures the traffic load that should be used in further studies to achieve saturation conditions. This scenario consists of a single BSS containing one station and one AP, which communicate in the downlink direction over one or two antennas, as presented in Figure 4.1. The simulation parameters are presented in Table 4.4.

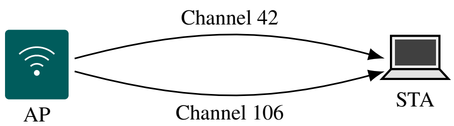


Fig. 4.1. Topology of throughput scenario 1: a single BSS in which one AP serves one station in the downlink direction over one or two 5 GHz links (80 or 160 MHz wide).

Table 4.4. Simulation parameters for throughput scenario 1.

Simulation parameter	**Value**
Band	5 GHz
Channels	42, 106 (80 MHz); 50, 114 (160 MHz)
Channel width	80, 160 MHz
Number of links	1, 2
MLO mode	STR
MCS	11
A-MPDU size	64, 1024 MPDUs
AP–STA distance	1 m

Figure 4.2 shows that, for a single-link configuration with maximum frame aggregation (A-MPDU = 1024 MPDUs), the throughput on a 160 MHz channel is nearly double that on an 80 MHz channel. A higher A-MPDU aggregation mode brings a slight gain, but the real difference appears with two links: more MPDU aggregated frames provide more than double the performance for STR on two links. Similarly, the 160 MHz channel width doubles the performance, as expected. The results of the cited work show a similar tendency as presented for 1024 MPDU frame aggregation: a 160 MHz channel leads to nearly double the throughput as for one link, while for two links with MLO the performance is again doubled compared to a single link. The authors of [19] do not specify which aggregation mode they use; however, the characteristics of the results better fit the 1024 MPDU mode. Both studies confirm the expected behavior: the wider channel and the higher aggregation of A-MPDU frames, together with MLO operating in STR mode, significantly elevate the achievable Wi-Fi throughput.

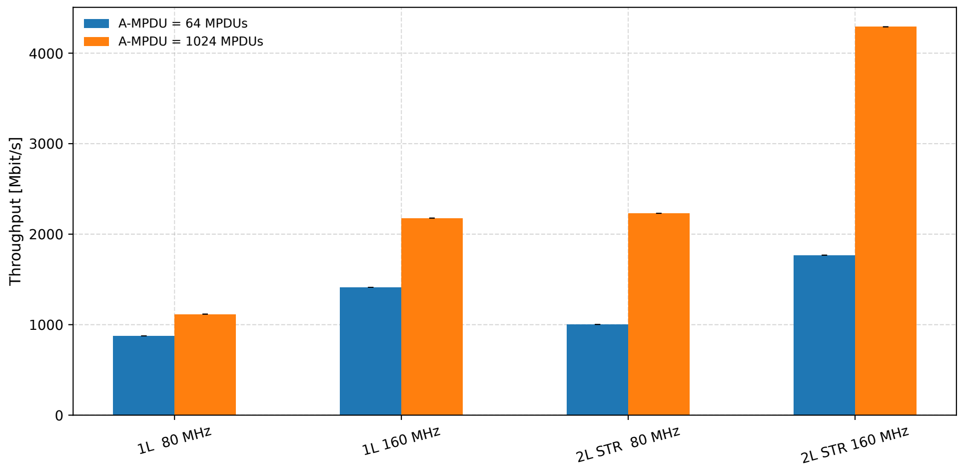


Fig. 4.2. Maximum throughput for different channel widths and link configurations.

A quantitative comparison with the cited work confirms these tendencies. The achievable throughput reported there (Table II in [19]) is about 450 Mb/s on a single 80 MHz link and 850 Mb/s on two such links, a gain of roughly 90%, and about 750 Mb/s on a single 160 MHz link against 1400 Mb/s on two, a gain of roughly 85%; in both cases two 80 MHz links also slightly exceed a single 160 MHz channel. The same three patterns appear in the present results – the near-doubling of the throughput with a second STR link, the doubling with the wider channel, and the small advantage of two 80 MHz links over a single 160 MHz channel – but only for the high aggregation mode (A-MPDU = 1024), which reproduces them closely, whereas the low aggregation mode (A-MPDU = 64) falls well short of doubling. The absolute throughput measured here is higher than in the cited work, which is consistent with the aggregation setting being left unspecified there and with the differences between the two simulation environments.

4.2.2. Throughput Scenario 2

The second scenario, inspired by [17], shows how the different MLO modes impact maximum performance. Two modes of MLO (STR and EMLSR) are compared with SLO. The topology, graphically described in Figure 4.3, consists of one AP and one station communicating in the downlink direction, using between one and three antennas. The simulation parameters are shown in Table 4.5.

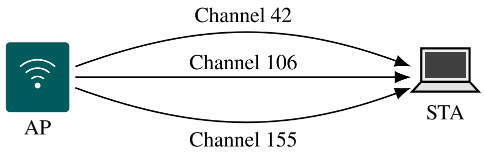


Fig. 4.3. Topology of throughput scenario 2: one AP communicates with one station in the downlink direction over one, two, or three 5 GHz links, depending on the transmission mode.

Table 4.5. Simulation parameters for throughput scenario 2.

Simulation parameter	**Value**
Band	5 GHz
Channels	42, 106, 155
Channel width	80 MHz
Number of links	1, 2 or 3
Number of stations	1
MLO mode	STR, EMLSR
MCS	8

Figure 4.4 shows that EMLSR performs nearly the same as SLO, having only a minimal decrease in performance due to channel switching. However, STR outperforms SLO by nearly a factor of two when using two antennas and by approximately 2.8 times when using three antennas. The benefit of using STR is clearly visible, substantially outperforming EMLSR in a basic scenario without additional traffic. Clearly, EMLSR is not designed for this case: it operates only on one link at a time, and the additional switching time between two links slightly reduces its performance compared to SLO.

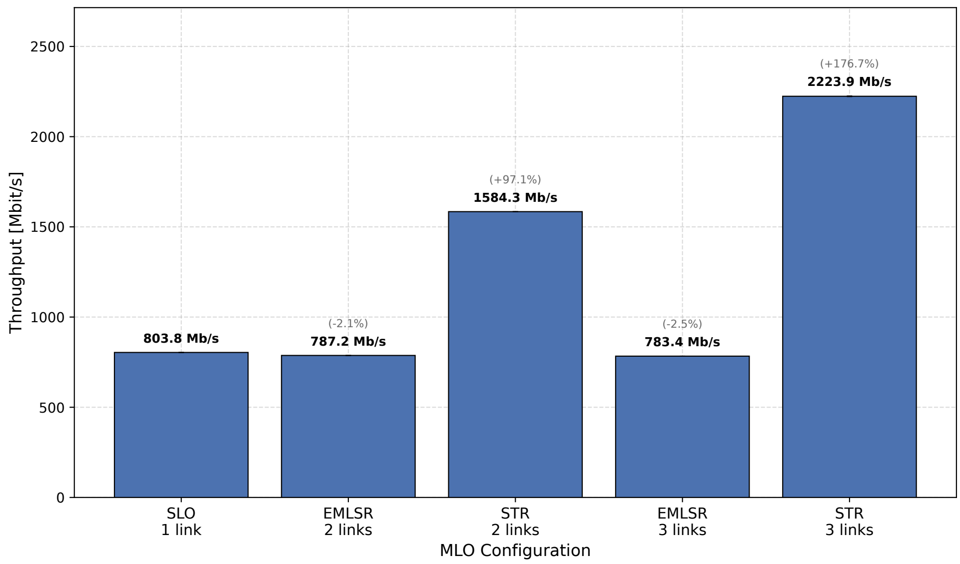


Fig. 4.4. Throughput values for different transmission modes and link number configurations. The percentages in grey refer to the change with respect to the SLO performance. The 95% confidence intervals are too small to be graphically visible.

The cited study [17] presents similar results: MLSR behaves in a similar way to a single-link connection, and STR's throughput is nearly doubled and tripled for two and three link connections, respectively. These are expected due to the principles of operation in both scenarios: EMLSR can send and receive data only on one link at a time, while STR can communicate using many links in parallel. However, the values differ, as they are 20% higher in this study than in the cited work. The discrepancies can originate from the slight differences in parameters, as in the cited study a list of the adjusted parameters is more detailed.

4.2.3. Throughput Scenario 3

The third scenario is also based on [17]. The goal is to discover how MLO operates in multiple overlapping BSSs, considering coexistence with SLO devices. The topology consists of three BSSs, each with one station and one AP communicating in the downlink direction. Two of the BSSs (BSS0 and BSS1) contain SLO devices operating on separate 5 GHz channels (42 and 106) and one (BSS2) includes MLO devices operating on both channels simultaneously. This specific configuration leads to competition for airtime between SLO and MLO devices, which can potentially worsen performance, especially throughput. The particular simulation parameters are presented in Table 4.6.

Table 4.6. Simulation parameters for throughput scenario 3.

Simulation parameter	Value
Band	5 GHz
Channels	42, 106
Channel width	80 MHz
Number of links	1, 2
Number of BSSs	3
MLO mode	STR, EMLSR
MCS	8

Figure 4.5 shows that for the STR transmission mode the throughput is higher than for the SLO devices, having nearly twice as much throughput as in both SLO cases. However, the EMLSR mode does not give any performance boost, having a slightly lower overall throughput than the SLO stations. This is expected due to the ability of STR to transmit over many links at the time, where EMLSR can only have one active transmitting link.

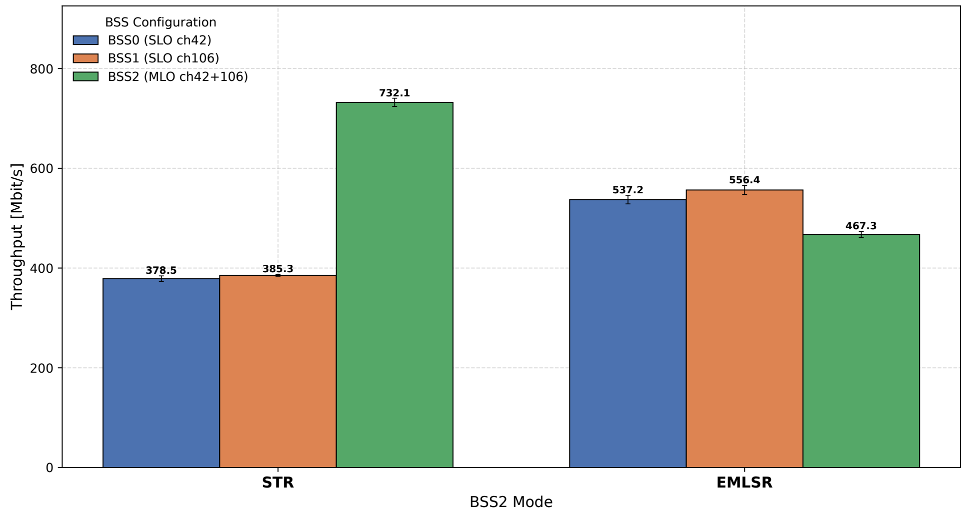


Fig. 4.5. Throughput results for MLO competing with two other BSSs.

In the cited work, the authors achieve the same result for MLSR as in the EMLSR mode, which matches expectations: the characteristics of EMLSR do not give a way to outperform SLO maximum throughput due to its possibility to transmit only on one link at a time. However, if one of the channels is highly occupied, EMLSR should give better results providing the option to switch to the other channel. There is also a small decrease in throughput induced by channel switching. Meanwhile STR's gain in throughput elicits an impact on the SLO devices, forcing them to compete for medium access, thus worsening their performance compared to EMLSR. As before, these results show approximately 20% higher throughput values than [17].

4.2.4. Throughput Scenario 4

The fourth scenario is inspired by [5]. The goal is to examine how SLO and different MLO modes behave under different BSS loads. The setup consists of one BSS containing one AP and one station communicating in the downlink or uplink directions over one or two links, and two BSSs in the role of serving background traffic, called the overlapping BSS (OBSS). The normalized load in these BSSs is set up symmetrically, which means that there is no discrepancy in load between links. Detailed simulation parameters are given in Table 4.7.

Table 4.7. Simulation parameters for throughput scenario 4.

Simulation parameter	Value
Band	5 & 6 GHz
Channels	5 GHz: 42, 6 GHz: 7
Channel width	80 MHz
Number of links	1, 2
Number of stations (main BSS)	1
Number of stations (OBSS)	1–10
MLO mode	STR, EMLSR
MCS	13

Figure 4.6 shows that the STR mode has the highest throughput in the first half of the measured cases. EMLSR initially has similar performance as SLO but with higher OBSS load EMLSR performs better than SLO. Note that within the 40-60% load interval, EMLSR has higher downlink throughput than STR, while in the majority of cases STR performs similarly or better. With loads greater than 60%, all transmission modes behave similarly with a slight predominance of STR.

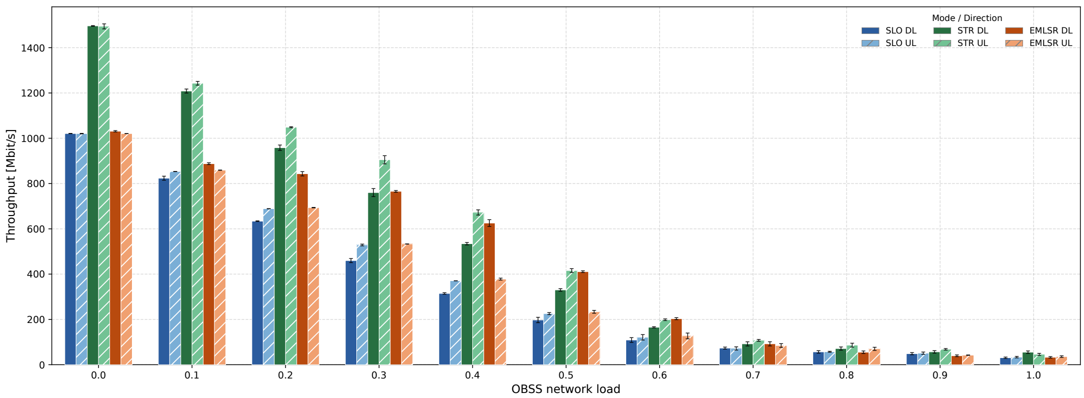


Fig. 4.6. Throughput of the different transmission modes under normalized OBSS load.

In the referenced work, the authors achieve the following results: STR outperforms other modes in all load configurations, near 50% load STR and EMLSR performance appears to be roughly the same. Upon 60% load, all three modes perform similarly. These observations are similar to Figure 4.6, with the exception that the STR mode is not twice as good as STR with 0% OBSS load. Considering the lower MCS settings, STR performance meets the ratio presented in the paper.

4.2.5. Throughput Scenario 5

The final throughput study is based on [31]. The purpose is to show how MPDU aggregation impacts throughput. The networks contains one AP and one station communicating in the downlink direction under a full buffer load. The communication is carried out on two 5 GHz links, one of which is subject to a variable load between 10 and 70%, and the second has a load of 0 or 10%. The detailed configuration of this scenario is presented in Table 4.8.

Table 4.8. Simulation parameters for throughput scenario 5.

Simulation parameter	Value
Band	5 GHz
Channels	42, 108
Channel width	80 MHz
Number of links	2
Number of stations	1
MLO mode	STR
MCS	11
A-MPDU size	64, 512 or 1024 MPDUs

Figure 4.7 presents the results of the throughput measurement for three different settings of MPDU aggregation, with two different TXOP limit values (0 and 3 ms). For both TXOP settings, A-MPDU = 64 MPDUs performs much worse than the two other values (512 and 1024), which give similar results. On the other hand, a 3 ms TXOP limit brings a slight gain in all of the results, but the main principle persists the same: an A-MPDU setting of 64 MPDUs gives worse results than higher aggregation modes. The throughput decreases in both cases with increasing load, which is expected due to increased contention caused by more transmitting stations.

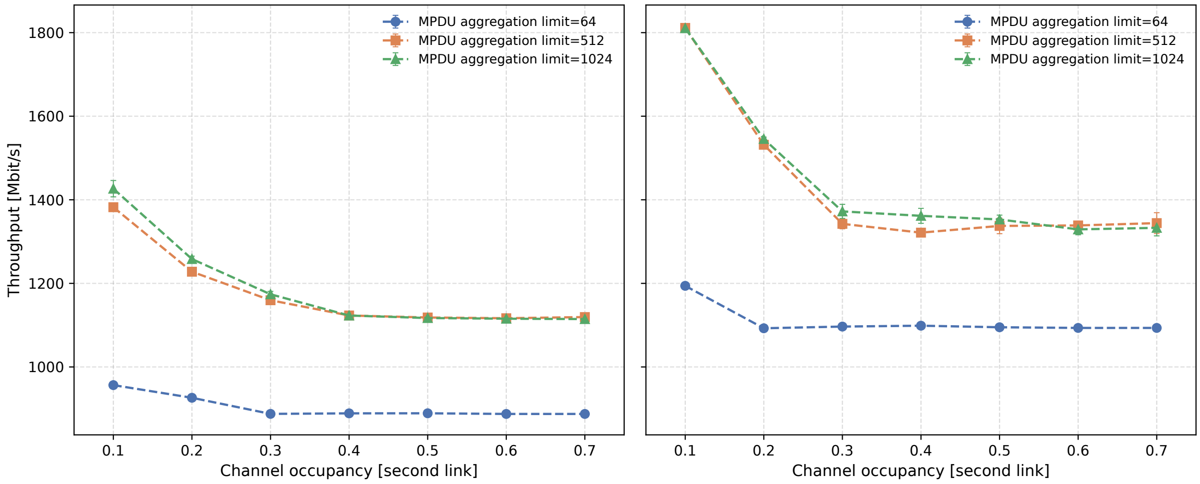


Fig. 4.7. Throughput results with a TXOP limit of 0 ms (left) and 3 ms (right) and without load on the first link.

Figure 4.8 shows that the 10% load on the first link has a meaningful impact on throughput, with the values starting from less than 1200 Mbit/s for the TXOP limit = 0 ms, compared to the 1400 Mbit/s starting point in Figure 4.7. As in the previous situation, changing the TXOP limit to 3 ms gives a slight bonus in performance, moving the starting point near 1500 Mbit/s. The decrease in throughput with increasing load is also present and expected.

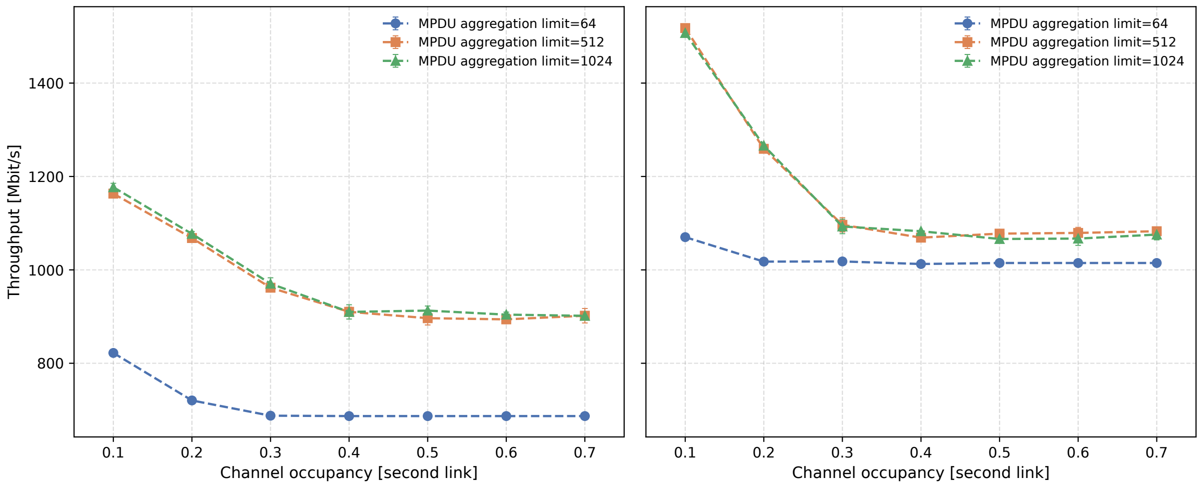


Fig. 4.8. Throughput results with a TXOP limit of 0 ms (left) and 3 ms (right) and with 10% load on the first link.

The cited work differs in some areas, such as the asymmetry of offered load proposed therein, but the core assumptions are similar. The difference between TXOP limit values is noticeable in favor of a 3 ms TXOP limit, but the biggest difference is the throughput achieved by the 512 and 1024 A-MPDU modes, where the lower aggregate value performs much better. Also, authors used a much larger offered load and did not describe their testbed precisely, so many parameters had to be assumed.

4.3. Latency

This section reports a number of scenarios showing how MLO impacts channel access delay. The ability to utilize more than one link and switch between links when the contention is higher on one of the channels is the biggest advantage of MLO. The following scenarios investigate how MLO impacts latency and whether the benefits are observable.

4.3.1. Latency Scenario 1

The first latency scenario is based on the work performed in [19]. The goal is to verify how the number of active stations and the offered load impact MLO operation and channel access delay. The scenario consists of a BSS containing one AP and a varying number of stations. One station is the focus of the latency study, the rest serve as background traffic. The traffic load is based on the results from the first throughput scenario (Section 4.2.1), with the load normalized to the maximum value reached in that study. In scenarios involving more than one user, the load is evenly distributed between stations. Detailed simulation parameters are shown in Table 4.9.

Table 4.9. Simulation parameters for latency scenario 1.

Simulation parameter	Value
Band	5 GHz
Channels	42, 106 (80 MHz); 50, 114 (160 MHz)
Channel width	80 MHz, 160 MHz
Number of links	1, 2
Number of stations	1, 4, 10
MLO mode	STR
MCS	11
A-MPDU size	64 MPDUs

Figure 4.9 presents the mean downlink latency for the 80 MHz channel width, comparing SLO and MLO STR for a varying number of stations. The results show that the latency generally increases with the normalized traffic load for all cases. This behavior is expected because a higher offered load causes more frequent channel access attempts, longer contention periods, and a higher channel access delay. The single link configuration with 10 users performs the worst in all cases, reaching nearly 5.6 ms for the highest load values. MLO STR achieves better results than all of the corresponding single link configurations, demonstrating a much slower latency growth.

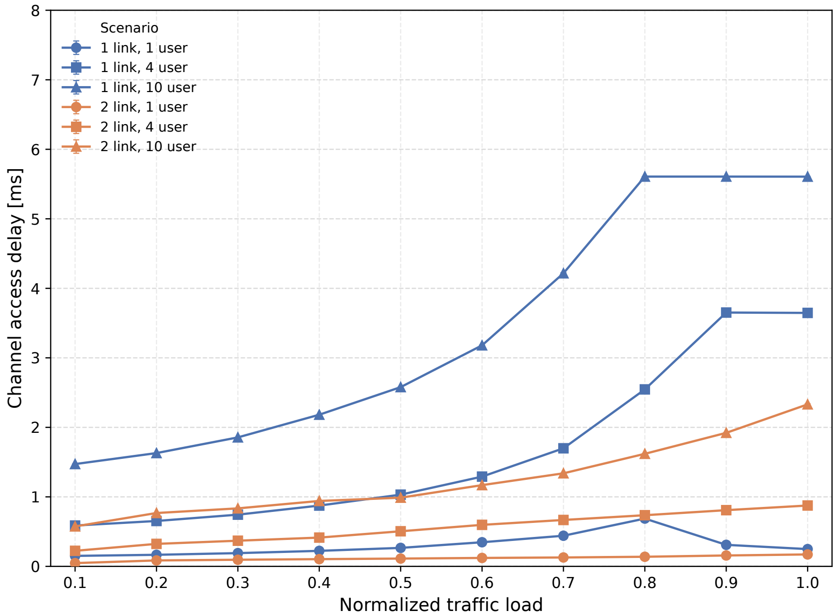


Fig. 4.9. Mean latency for 80 MHz channel and different load and user configurations.

Figure 4.10 shows the mean downlink latency for the 160 MHz channel width. As in the previous case, the latency generally increases with the offered traffic load and with the number of active users. Once again, all MLO configurations perform better than their SLO representatives. The single link case with 10 users has the highest maximum delay reaching around 6.8 ms, which is more than in the 80 MHz configuration. The same behavior can be observed for all configurations, having a maximum delay higher than in the 80 MHz case.

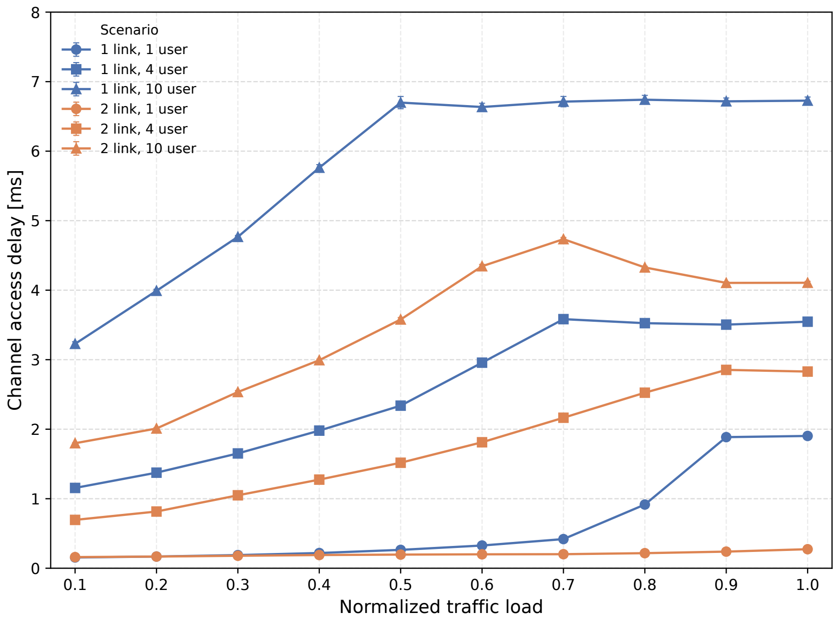


Fig. 4.10. Mean latency for 160 MHz channel and different load and user configurations.

Several latency fluctuations can be observed, especially near full-load conditions, where the relationship between traffic load and delay does not strictly follow the general trend. This behavior can be explained by the probabilistic nature of channel access and the operation of heavily loaded channels. Under such conditions, latency becomes more sensitive to random backoff values, collisions, retransmissions, contention window changes, and frame aggregation. Therefore, small deviations are expected and should not be interpreted as a contradiction to the overall trend. The method used to measure channel access delay in ns-3 is not precise and can be the source of these artifacts. The confidence intervals are not clearly visible on the plots as a result of the high precision and repeatability of the simulation results.

Compared with the referenced work, the general behavior of the results is similar: SLO latency increases rapidly as the number of users and the offered load grows, while MLO STR keeps the delay at a lower level. The exact values are close to the referenced results, especially the maximum values reached by the MLO STR.

4.3.2. Latency Scenario 2

The second latency scenario is taken from [18]. The objective is to evaluate how different traffic loads affect latency in a contention-free environment, comparing MLO STR using two or four links with SLO. The scenario consists of one AP and one station communicating in the downlink direction. In the SLO case, the devices operate on a single link in the 5 GHz band, while in the MLO case they communicate over two or four links spread across the 5 and 6 GHz bands, each with a channel width of 80 MHz. The offered load is swept across a wide range to gradually drive the link from low occupancy towards saturation. Detailed simulation parameters are shown in Table 4.10.

Table 4.10. Simulation parameters for latency scenario 2.

Simulation parameter	**Value**
Band	5 & 6 GHz
Channels	42, 108 (5GHz); 7, 23 (6 GHz)
Channel width	80 MHz
Number of links	1, 2 or 4
Number of stations	1
MLO mode	STR
MCS	8
A-MPDU size	1024 MPDUs

Figure 4.11 presents the 50th and 99th percentiles of downlink latency as a function of the offered load. For all configurations, the latency remains low and nearly flat while the link is unsaturated, and then rises sharply once the offered load approaches the capacity of the available links. The single link configuration is the first to saturate, with its latency growing steeply at the lowest loads. Adding links shifts this saturation point towards higher loads: the two-link MLO STR keeps the delay low over a substantially

wider range, and the four-link sustains low latency up to the highest offered loads. The gap between the 50th and 99th percentiles widens as saturation is approached, reflecting the growing influence of queueing and the high maximum contention window values achievable under heavy load.

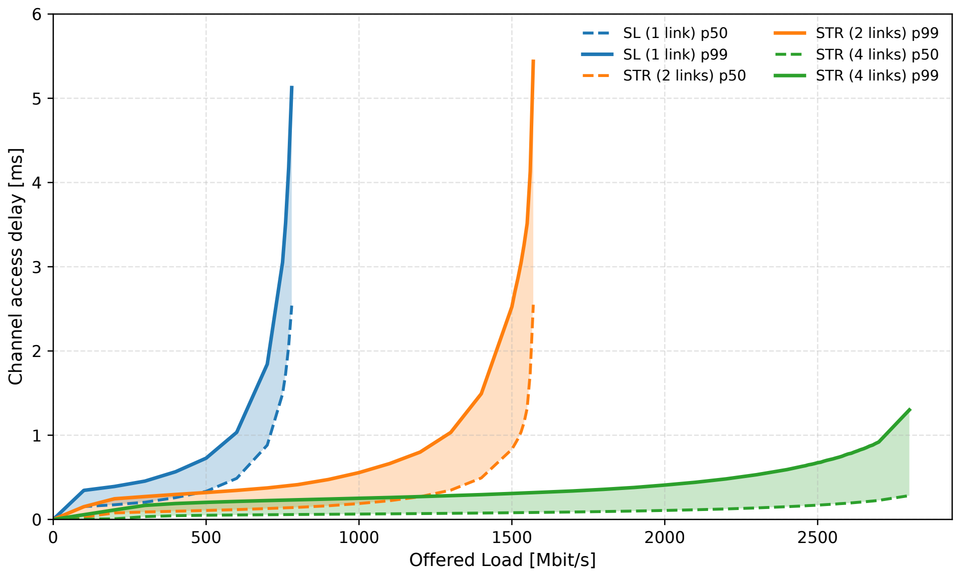


Fig. 4.11. Downlink latency percentiles (50th and 99th) versus offered traffic load for SLO and MLO STR (1, 2, and 4 links).

4.3.3. Latency Scenario 3

Based on [17], the third latency scenario examines how traffic load and number of active links jointly influence latency of SLO and MLO (in STR and EMLSR modes), again in a contention-free environment. The topology involves a single AP serving one station in the downlink direction over one, two or three links, each 80 MHz wide in the 5 GHz band. The MLO devices operate in either STR or EMLSR mode using two or three links, while the offered load is normalized to the maximum value reached for the corresponding configuration. The detailed simulation parameters are summarized in Table 4.11.

Table 4.11. Simulation parameters for latency scenario 3.

Simulation parameter	**Value**
Band	5 GHz
Channels	42, 106, 138
Channel width	80 MHz
Number of links	1, 2 or 3
Number of stations	1
MLO mode	STR, EMLSR
MCS	8
A-MPDU size	1024 MPDUs

Figure 4.12 reports the average and the 99th percentile latency for the SLO, EMLSR and STR modes using two or three links as a function of the normalized traffic load. As expected, the latency grows with the offered load for all configurations. The STR mode significantly reduces latency compared to the SLO, as it is able to transmit over multiple links simultaneously; in a contention-free scenario this capability can be fully exploited, leading to visible improvements at both low and high traffic loads. The EMLSR mode, on the other hand, does not bring such a benefit, since it can be active on only one link at a time, and the additional link switching introduces extra delay.

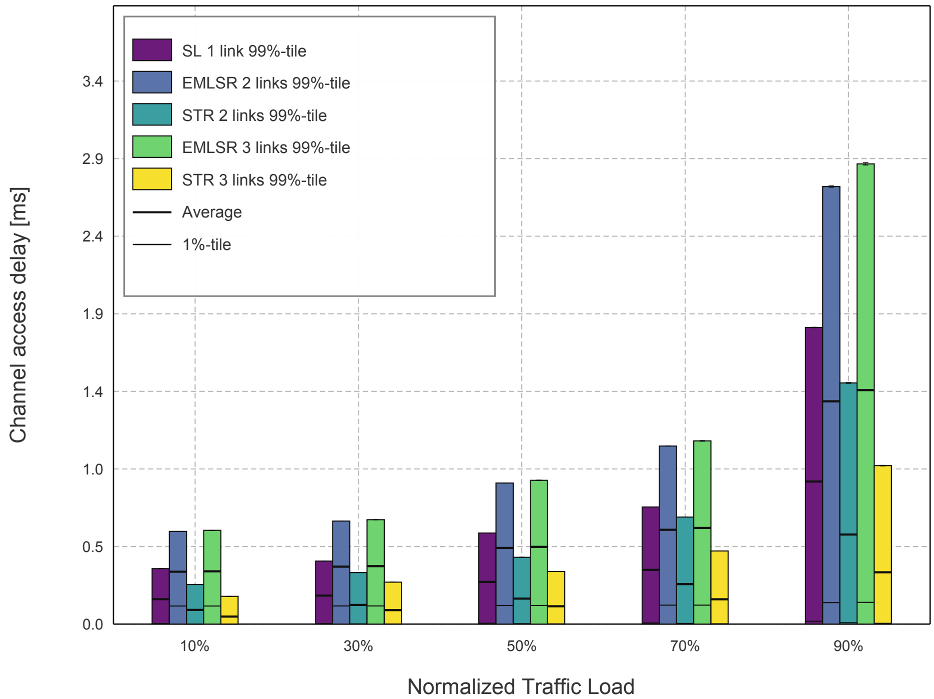
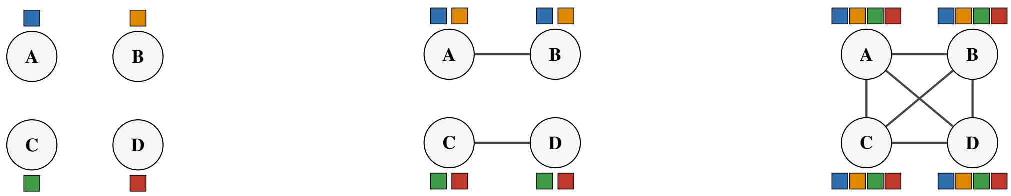


Fig. 4.12. Average and 99th percentile downlink latency versus normalized traffic load for SLO, MLO STR, and MLO EMLSR (2 and 3 links).

In the cited work, STR behaves consistently with the results presented here, reducing latency relative to SLO across the whole load range. EMLSR, however, does not outperform SLO: this behavior can be attributed to the additional protocol overhead and the link switching time inherent to this mode. These observations confirm that in contention-free conditions only the simultaneous transmission offered by STR translates into a clear latency reduction.

4.3.4. Latency Scenario 4

Taken again from [18], the fourth scenario moves to a contended setting to examine the impact of traffic load on latency across four overlapping BSSs and to look for the latency anomaly reported by the authors, where SLO can paradoxically achieve lower latency then MLO.



(a) SLO: one private channel per BSS.

(b) MLO-STR:2: two pairs share two channels.

(c) MLO-STR:4: all four channels shared.

Fig. 4.13. Topology of latency scenario 4: four overlapping BSSs (A–D) in the three operation modes. Each colour denotes one channel (blue: 42, orange: 106 in 5 GHz; green: 7, red: 23 in 6 GHz); an edge joins two BSSs that share a channel.

As presented in Figure 4.13, each of the four BSSs hosts one AP and one station exchanging downlink traffic. Three modes of operation are considered: SLO, in which each BSS operates on a single, non-shared 80 MHz channel; MLO STR with two links, where there are two pairs of BSSs sharing the same two channels; and MLO STR with four links, in which all four channels are shared among all of the BSSs. The detailed configuration is given in Table 4.12.

Table 4.12. Simulation parameters for latency scenario 4.

Simulation parameter	**Value**
Band	5 & 6 GHz
Channels	42, 106 (5 GHz); 7, 23 (6 GHz)
Channel width	80 MHz
Number of links	1, 2 or 4
Number of BSSs	4
MLO mode	STR
MCS	8

Figure 4.14 shows the 50th and 99th percentile downlink latency for SLO, MLO STR with two and four links as a function of the total offered traffic load. The median latency improves only slightly when multiple links are used, while the 99th percentile latency increases with the number of links across all traffic loads. This behavior is consistent with the latency anomaly described in the cited work: STR stations accessing multiple links block their contending neighbors, so adding links does not reduce latency, since each additional link introduces another channel on which contention can occur.

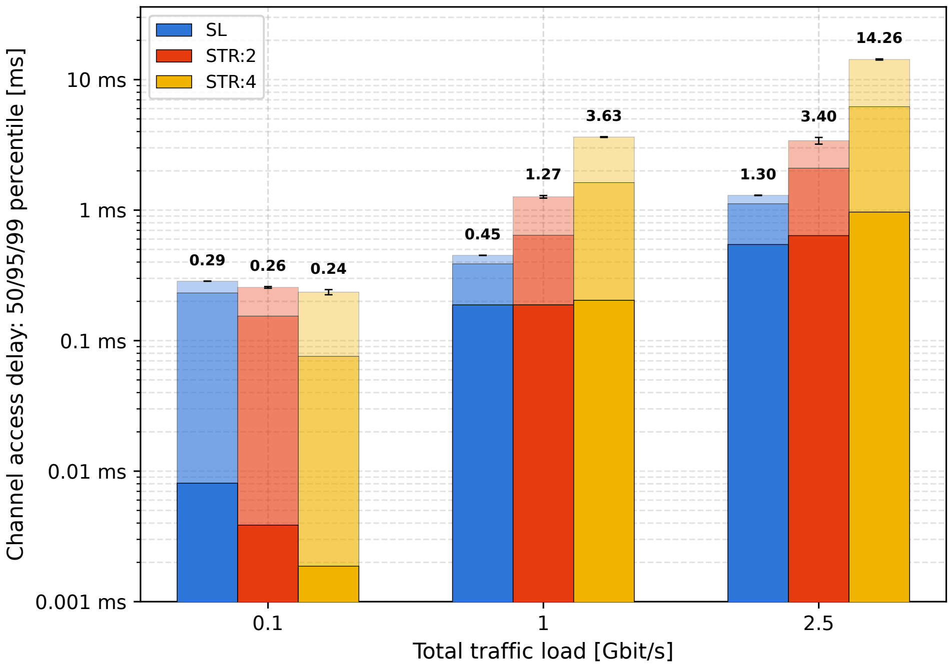
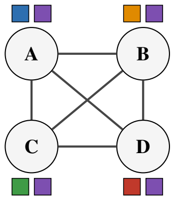


Fig. 4.14. Downlink latency percentiles in four overlapping BSSs versus total offered traffic load for SLO and MLO STR (2 and 4 shared links).

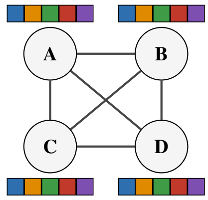
Compared with the referenced work, the same latency anomaly is observed: increasing the number of shared links does not lower the tail latency and may even increase it. With a low traffic load, adding more links is beneficial for latency, but with higher loads this tendency turns around, leading to higher latency paired with additional links. The study confirms the theory that in a contended environment, the simultaneous access of STR stations to several channels comes at the cost of higher delays for competing BSSs.

4.3.5. Latency Scenario 5

Continuing the four-BSS contention setup of the previous scenario and still following [18], the fifth scenario explores link-assignment strategies aimed at overcoming the latency anomaly identified above. The topology, presented in Figure 4.15, again places one AP and one station in each of the four BSSs, all transmitting in the downlink direction. Besides SLO and MLO STR with two links, three further configurations are studied: MLO EMLSR where each BSS uses two links shared with two other SLO BSSs; MLO STR with two links, where one link uses a shared channel among all four BSSs and another utilizes an exclusive channel; and MLO STR with five links, where every BSS uses the same five channels. The parameters of this configuration are collected in Table 4.13.



(a) MLO-STR:1+1: one private channel per BSS plus one channel (violet) shared by all.



(b) MLO-STR:5: all five channels shared among the four BSSs.

Fig. 4.15. Topology of latency scenario 5: the two link-assignment strategies across four BSSs (A–D). Each colour denotes one channel (blue: 42, orange: 106 in 5 GHz; green: 7, red: 23, violet: 39 in 6 GHz); an edge joins two BSSs that share a channel.

Table 4.13. Simulation parameters for latency scenario 5.

Simulation parameter	Value
Band	5 & 6 GHz
Channels	42, 106 (5 GHz); 7, 23, 39 (6 GHz)
Channel width	80 MHz
Number of links	2 or 5
Number of BSSs	4
MLO mode	STR, EMLSR
MCS	8

Figure 4.16 compares the channel access delay of the additional modes against the SLO and MLO STR with two links, reported as 50th and 99th percentiles at three representative total loads. At the lowest load (100 Mbit/s) the medium is essentially uncongested and all modes behave alike, with a delay well below a millisecond. As the load grows, the median stays low for every mode – it never exceeds about 0.65 ms even at 2500 Mbit/s – so the modes are distinguished almost entirely by their 99th percentile, that is, by the highest achievable delay. The decisive factor turns out to be the way the links are assigned. MLO STR 1+1, in which each BSS keeps one private link, achieves the lowest tail at every load and, under heavy traffic, falls clearly below even SLO, confirming that a dedicated link protects a BSS from contention escalation. MLO STR and EMLSR with two links behave almost identically and sit in the middle, with a tail of roughly 3.6 ms at 2500 Mbit/s. The fully shared MLO STR where all five links are contended by the four BSSs, performs worst of all: its 99th percentile grows steeply with the load, reaching about 7 ms at 2500 Mbit/s, well above the SLO baseline.

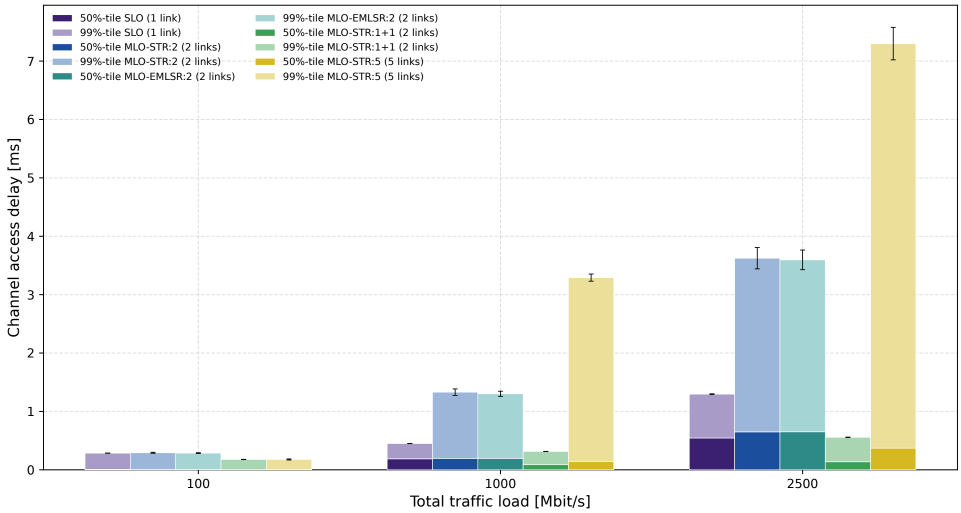


Fig. 4.16. Downlink latency percentiles (50th and 99th) versus total offered traffic load for link-assignment strategies in the four-BSS contended scenario.

These results only partly agree with [18]. The benefit of giving each BSS a private link is confirmed: MLO STR 1+1 is the best configuration and the only one that keeps the 99th percentile delay way below the SLO under heavy load. MLO STR with five links behaves in the opposite way to the reference – instead of ranking among the best, it reaches the highest maximum delay, because sharing all five links across the four BSSs multiplies the number of collisions through the channels. MLO EMLSR with two links behaves in a similar way to MLO STR, also with two links, while in the study MLO STR with two links visibly underperforms MLO EMLSR.

4.3.6. Latency Scenario 6

The final latency scenario, based on [20], turns to the effect of channel occupancy, assessing how different levels of background activity influence the latency of MLO STR devices relative to SLO, and whether adding a second, more heavily occupied link can still reduce delay. The setup, visually presented in Figure 4.17, centres on a main BSS in which one AP serves one station in the downlink direction over one or two 80 MHz links in the 5 GHz band, under full-buffer load. The occupancy is generated by two OBSSs, one per link, each populated with a variable number of single link stations. The two channels may be loaded either symmetrically or asymmetrically, and the single link devices always use the less congested channel. Table 4.14 summarizes the simulation parameters of this study.

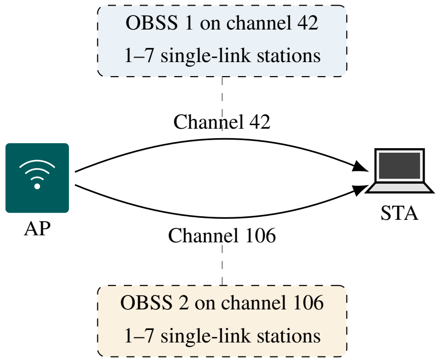
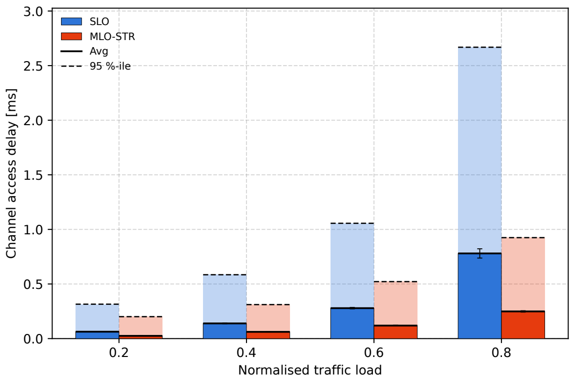


Fig. 4.17. Topology of latency scenario 6: a main BSS in which one AP serves one station in the downlink direction over one or two 5 GHz links; each link is loaded by a separate OBSS with a variable number of single-link stations that model the channel occupancy.

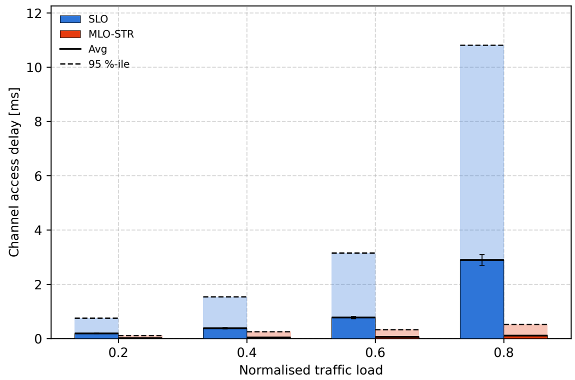
Table 4.14. Simulation parameters for latency scenario 6.

Simulation parameter	**Value**
Band	5 GHz
Channels	42, 106
Channel width	20 MHz
Number of links	1, 2
Number of stations (main BSS)	1
Number of stations (OBSS)	1, 4 or 7
MLO mode	STR
MCS	11

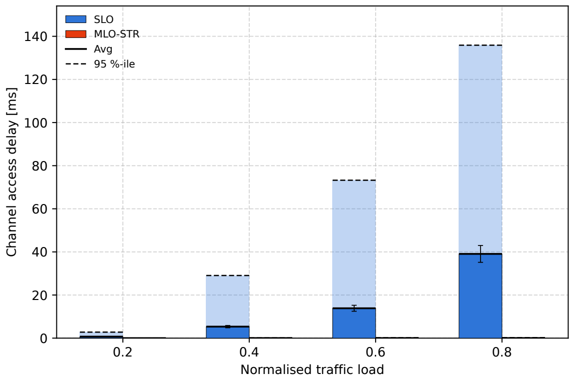
Figure 4.19 presents the average and 95th percentile downlink latency under symmetrical channel occupancy of 10%, 40%, and 70%, as a function of the normalized traffic load. In all three cases, the MLO STR channel access delay is lower than the SLO delay at every load, for both the average and the 95th percentile, and the gap widens as the load grows. The decisive factor is the occupancy level. The SLO delay, which is set by the single channel the station uses, escalates as that channel becomes busier: its 95th percentile at the highest load rises from about 2.7 ms at 10% occupancy to roughly 11 ms at 40% and to around 136 ms at 70%. MLO STR delay, in contrast, stays small throughout – below 1 ms at both 10% and 40% occupancy, and practically negligible at 70%. The heavier the symmetric occupancy, the more valuable the second link becomes, since it lets the MLO station sidestep the congestion that stresses the channel.



(a) 10% occupancy on both channels.



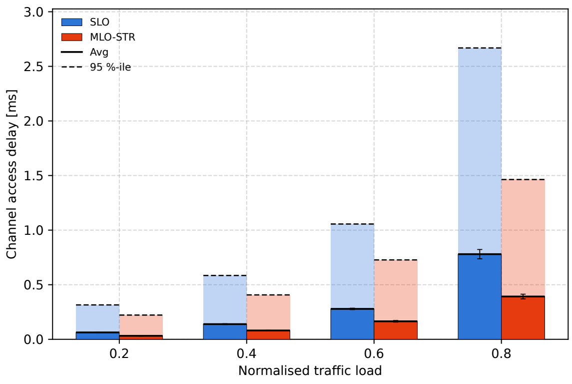
(b) 40% occupancy on both channels.



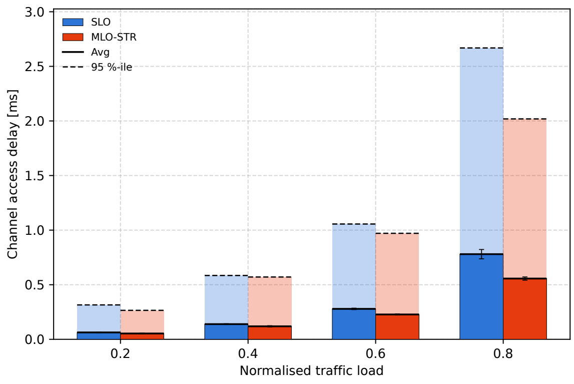
(c) 70% occupancy on both channels.

Fig. 4.18. Downlink latency versus normalized traffic load under symmetrically occupied channels.

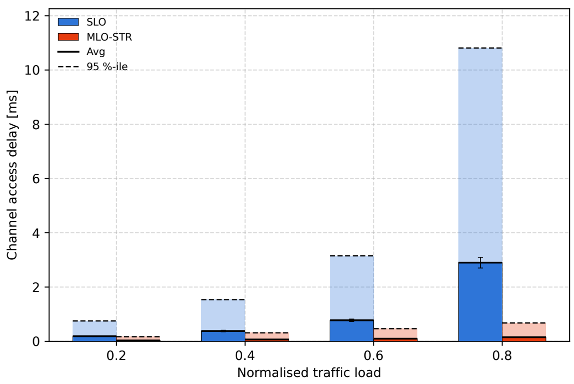
Figure 4.19 reports the same metric for asymmetrical occupancy, with the first and second channels loaded to 10%/40%, 10%/70%, and 40%/70% respectively. In every combination, MLO STR keeps a lower channel access delay than SLO over the whole load range, although the size of the advantage varies. For the 10%/40% pair, MLO stays comfortably below SLO, with the 95th percentile at about 1.5 ms against 2.7 ms at the highest load. For the 10%/70% pair, the margin is the narrowest: MLO and SLO are almost level at low and medium loads, and MLO pulls clearly ahead only near saturation (about 2.0 against 2.7 ms). The 40%/70% pair uncovers the widest gap of all, MLO having the highest delay at less than 1 ms while SLO increases to about 11 ms at the highest load. Because the MLO station can always fall back on its second link, its delay tracks the less congested channel rather than the busier one, so the second link remains beneficial and increasingly so as the occupancy grows (even when it is significantly busier than the first).



(a) 10% / 40% channel occupancy.



(b) 10% / 70% channel occupancy.



(c) 40% / 70% channel occupancy.

Fig. 4.19. Downlink latency versus normalized traffic load under asymmetrically occupied channels.

These results only partly match the trends reported in [20]. The advantage of MLO under symmetric occupancy is reproduced and is in fact pronounced. The latency anomaly described in that work, however, where committing a packet to a heavily occupied link before the backoff can make the extra link a liability and push the MLO delay above that of the SLO, does not appear in these simulations: across every tested occupancy combination, including the strongly asymmetric ones, the second link consistently lowers the channel access delay. The discrepancy is most likely due to differences between the simulators, especially the schedulers used.

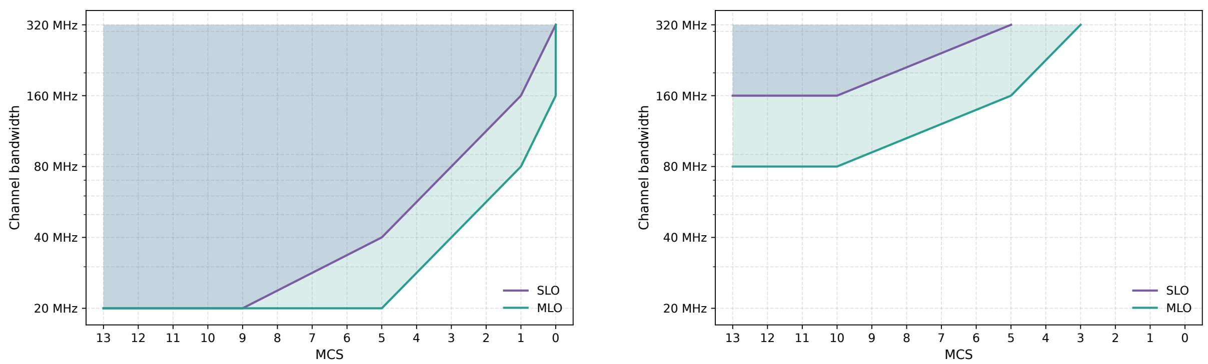
4.4. VR Scenario

VR streaming is among the most demanding services a Wi-Fi network can carry, as it combines a high bursty data rate with strict delay limits. The limits used as a reference here follow the Wi-Fi Alliance VR gaming guidelines [41], which require the downlink channel access delay to remain below 5 ms at the 75th percentile and the uplink delay below 2 ms at the 90th percentile. Adopting the VR traffic profile and scenario topology of [35], this scenario asks which physical layer settings are sufficient for VR and how much a second link helps in reaching them. A single station communicates with the AP in one direction at a time, downlink or uplink; in the MLO case, the station spreads its traffic over two STR links, while in the SLO case it remains on a single channel. Rather than fixing the transmission rate, the experiment sweeps many PHY settings (from MCS 0 to MCS 13 and from 20 MHz up to 320 MHz of channel width) and records where the delay limit is satisfied. The configuration is summarized in Table 4.15.

Table 4.15. Simulation parameters for the VR scenario.

Simulation parameter	Value
Band	6 GHz
Channels	29, 153 (20 MHz); 27, 155 (40 MHz); 23, 151 (80 MHz); 47, 175 (160 MHz); 31, 159 (320 MHz)
Channel widths	20, 40, 80, 160 or 320 MHz
Number of links	1, 2
Number of stations	1
MLO mode	STR
MCS	0–13

Figure 4.20 marks, for the downlink and the uplink, respectively, the MAC and channel width combinations that keep the latency within the required bound. Comparing the two subfigures reveals a clear difference in difficulty. The downlink limit is the easier one: a large part of the combinations qualify, and SLO already satisfies it once a moderate MCS or a wider channel is used. The uplink limit is much more restrictive, with its feasible region limited to combinations of high MCS and wide channels. In both directions, the area covered by MLO is larger than the one covered by SLO, which means that the same delay target is reached at a lower MCS or on a narrower channel when two links are available, as neither link has to carry the whole traffic on its own.



(a) Downlink (75th percentile ≤ 5 ms).

(b) Uplink (90th percentile ≤ 2 ms).

Fig. 4.20. VR scenario: required channel bandwidth to achieve Wi-Fi Alliance VR gaming guidelines [41].

The main observation is that, on Wi-Fi 7, VR is essentially an uplink-limited problem, and that MLO is most valuable precisely in the region where the deadline is hardest to meet. This is in line with the direction of the results in the cited study; the specific MCS and bandwidth thresholds differ, as can be expected from the different simulator and traffic implementation, but the relative ordering of the two operation types is preserved, with MLO outperforming SLO.

4.5. Coexistence with Legacy Stations

A real Wi-Fi 7 deployment will rarely consist of multi-link devices alone; it will have to share the spectrum with the large base of legacy SLO stations. Following [32], this scenario looks at what happens when the two populations compete for the same medium, and in particular at how the throughput is split between them as the mix shifts from mostly modern to mostly legacy devices. A single AP serves 50 stations scattered at random within a radius of 7.5 m, all sending uplink full-buffer traffic. The legacy stations are SLO devices restricted to the 2.4 GHz band whereas the MLO stations bond three links across the 2.4, 5 and 6 GHz bands and run in either STR or EMLSR mode. The legacy fraction is increased step by step from 10% to 90%. Table 4.16 lists the configuration used.

Table 4.16. Simulation parameters for the legacy coexistence scenario.

Simulation parameter	Value
Band	2.4 & 5 & 6 GHz
Channels	2.4 GHz: 6, 5 GHz: 42, 6 GHz: 7
Channel width	80 MHz
Number of links	1, 3
Number of stations	50
Legacy devices	10%–90%
MLO mode	STR, EMLSR
MCS	8
A-MPDU size	64 MPDUs

The throughput is reported from three complementary perspectives, in each case as a function of the share of legacy devices and for both MLO modes. The aggregated system throughput in Figure 4.21 separates the two modes sharply. With STR, the aggregate first grows as the legacy share increases, peaks at around 60% of legacy devices, and only collapses once the network becomes almost entirely legacy at 90%; the early increase reflects the lower mutual contention among the few remaining MLO stations, while the final drop appears when too few of them are left to exploit the 5 and 6 GHz links. With EMLSR the aggregate stays almost flat over the whole range and well below the STR curve, since being active on a single link at a time limits how much the mode can pull from the medium.

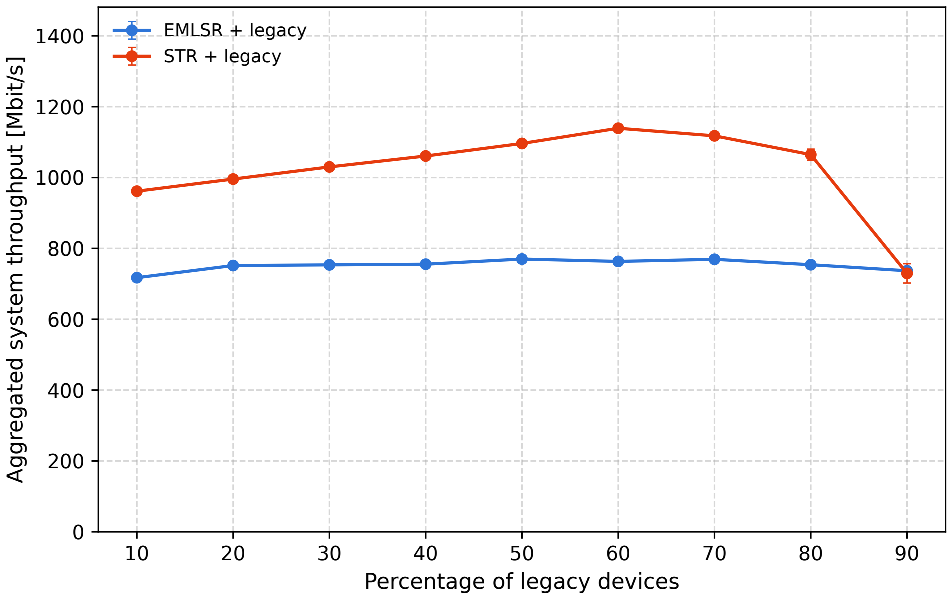


Fig. 4.21. Aggregated system throughput versus the percentage of legacy devices (STR and EMLSR).

Figure 4.22 splits the aggregate throughput between the two device classes. In both modes the multi-link devices account for the bulk of the throughput, whereas the legacy aggregate remains comparatively small. The two modes nonetheless treat the legacy class in opposite ways: under STR the legacy aggregate slowly grows with the number of legacy stations, while under EMLSR it starts higher but gradually declines, the two curves meeting only when legacy devices make up about 90% of the network.

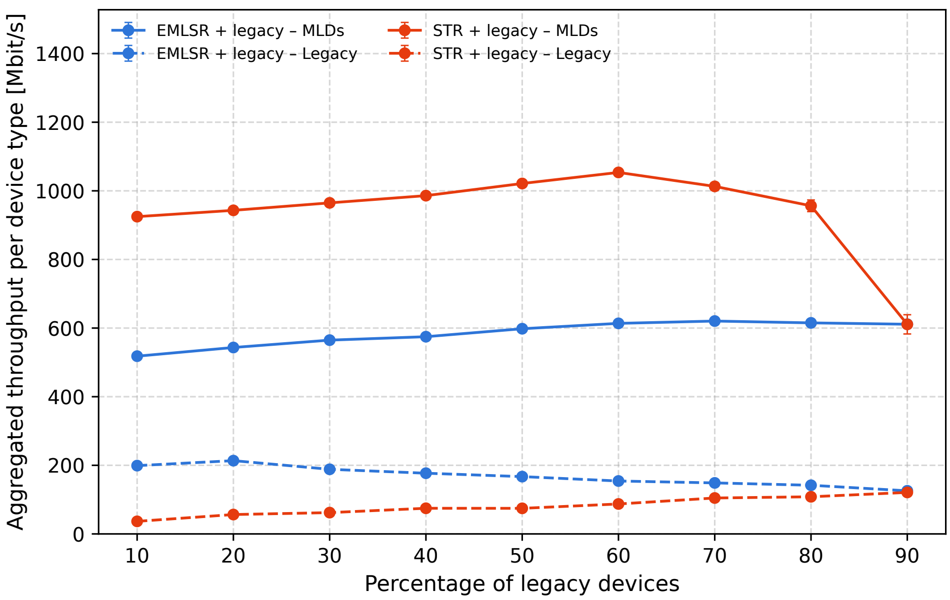


Fig. 4.22. Aggregated throughput per device type versus the percentage of legacy devices (MLO vs. legacy, STR and EMLSR).

The average throughput per device in Figure 4.23 makes the imbalance between the classes explicit. From the MLDs perspective the per-device throughput climbs steeply as their number shrinks, because a smaller group of MLDs shares the same multi-link capacity; the STR and EMLSR curves rise together and converge near 300 Mbit/s when the network is at 90% legacy. The legacy perspective follows the opposite trend: the per-device throughput is highest when only a handful of legacy devices are present, especially under EMLSR, which does not generate as much contention; as the legacy population grow, the throughput decays to a few Mbit/s.

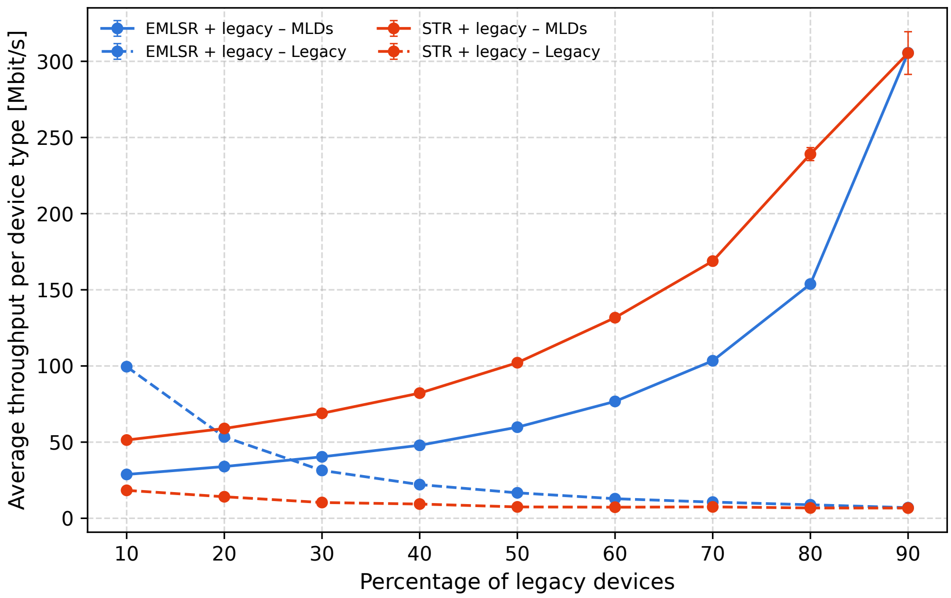


Fig. 4.23. Average throughput per device type versus the percentage of legacy devices (MLO vs. legacy, STR and EMLSR).

Compared with the reference [32], the per-device results agree on the essential point – MLO stations capture a disproportionate share of the medium, and the gap widens as their number increases. What differs is that aggregated throughput initially increases with a growing legacy population, starting to fail after raising the legacy number to 60%. A crucial point is that the study employs an unspecified MLO mode whose operating principle corresponds to single-link access. Consequently, STR may exhibit different performance compared to the reported results due to its ability to access multiple links simultaneously.

4.6. Comparison of Results

Taken together, the scenarios examined in this chapter characterize the conditions under which MLO gives performance gains and those under which it does not. This section consolidates the individual observations into a unified comparison, structured around the two metrics: throughput and latency.

For throughput, the two MLO modes behave differently, and the gain depends strongly on the scenario rather than being uniform. Figure 4.24 collects the throughput gain of each mode in SLO from all relevant throughput scenarios. STR improves on SLO in every case, but the largest gain (about 98%) appears in the contention free reference of Throughput Scenario 1, where the second link essentially doubles the available capacity; under contention, the relative gain shrinks to roughly 91% in Throughput Scenario 3 and 55% in Throughput Scenario 4, since the competing BSSs claim part of the medium as well. EMLSR is far more sensitive to the setting: it loses about 15% against SLO in Throughput Scenario 3 and only turns into a modest 20% gain in Throughput Scenario 4. This confirms that EMLSR, restricted to one active link at a time, pays off only when contention makes the ability to pick a less busy link worthwhile, and may otherwise fall behind plain SLO.

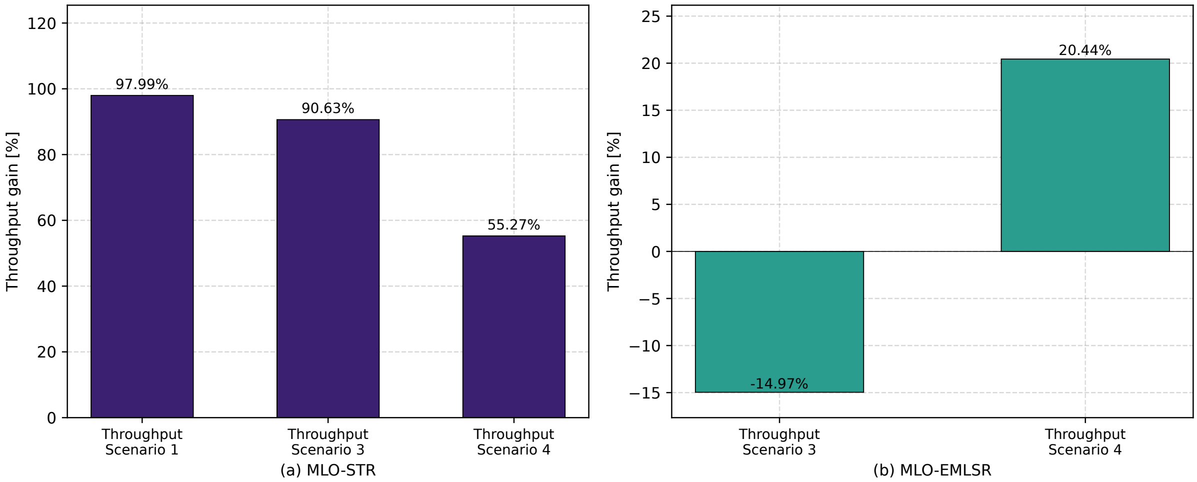


Fig. 4.24. Throughput gain of MLO over SLO across selected throughput scenarios.

On the latency side, the results split clearly between contention-free and contention-based conditions. In the contention free latency scenarios, gathered in Figure 4.25, MLO consistently and substantially lowers delay: the reduction is about 62% in Latency Scenario 1 and climbs to almost 100% in Latency Scenarios 2 and 3 (about 99.6% and 93%), where the spare links act as pure additional capacity. Under contention the outcome is no longer uniform, as shown in Figure 4.26, in which a positive value marks a delay increase with respect to SLO and a negative one – a reduction. The MLO configurations make things worse: the delay grows by about 66% in Latency Scenario 4 and by 34% for the MLO EMLSR variant with two shared links originating from the Latency Scenario 5: each additional shared link introduces a contention possibility. The same scenario gives completely different results for the MLO STR 1+1 configuration, where one link has an uncontended channel: the delay is lowered by 54%. Meanwhile, Latency Scenario 6 brings a 99% reduction of the delay, due to the low contention on the second channel on which MLO can switch the transmission (asymmetric scenario) and the abilty of balancing the traffic between links (symmetric scenario). The benefit of using MLO is therefore conditional, depending on both the offered load and the channel-allocation policy rather than on the number of the links alone. These results also highlight the importance of scheduling in MLO: for low-latency applications, frames should be assigned to queues on links that can provide a low channel-access delay, rather than distributed across links in a round-robin fashion.

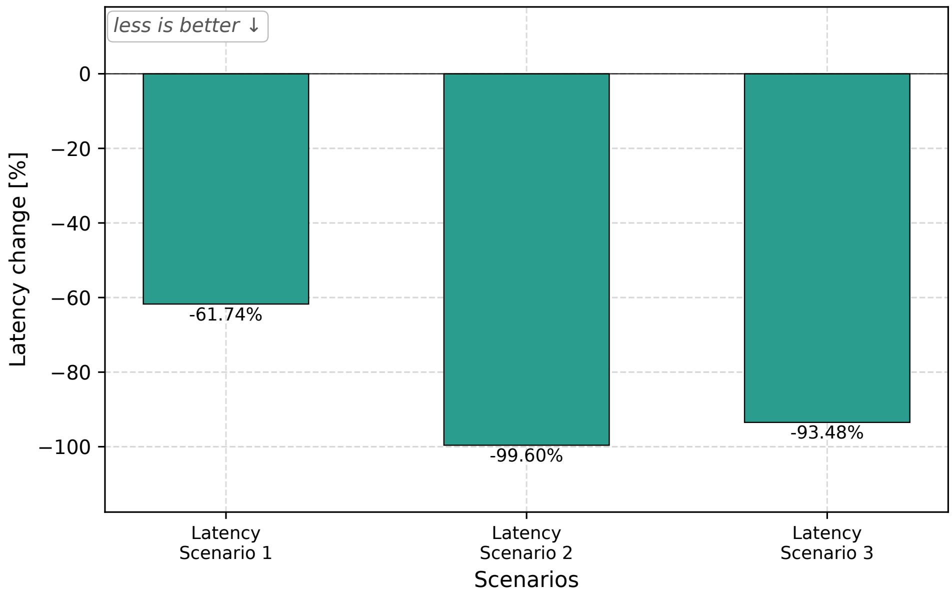


Fig. 4.25. Latency gain of MLO over SLO in the uncontended latency scenarios.

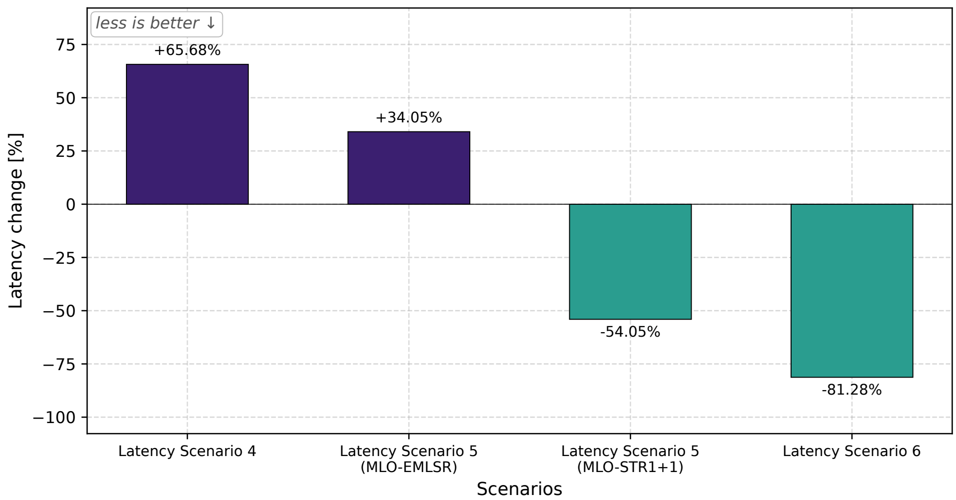


Fig. 4.26. Latency gain of MLO over SLO in the contended latency scenarios.

Finally, Table 4.17 consolidates these reproductions, listing for each scenario the figure or table in the cited work, the corresponding figure in this thesis, and how the present results compare with the source. The table shows that across the thirteen reproductions the simulations recover the qualitative behaviour reported in the literature: the throughput scaling of STR with the number of links, the near-irrelevance of EMLSR in the absence of contention, the contention-free latency reductions, and the channel-access anomalies under shared links all appear in the same direction as in the cited studies. The agreement is for the most part quantitative as well.

Table 4.17. Comparison between the reproduced scenarios and the corresponding results reported in the source studies.

Source	**Source fig./tab.**	**Figure(s) in this thesis**	**Comparison with the source study**
[19]	Tab. II	4.2	Same three trends recovered, although only for high A-MPDU. The absolute throughput obtained here is higher, which is consistent with the aggregation setting being left unspecified in the source.
[17]	Fig. 3	4.4	Matches the source: STR scales with the number of links, reaching approximately 2× and 2.8× SLO throughput, while EMLSR remains close to SLO.
[17]	Fig. 6	4.5	Agrees with the source: under two-BSS coexistence, STR roughly doubles SLO throughput, whereas EMLSR provides no clear throughput gain.
[5]	Fig. 5	4.6	Reproduces the ordering reported in the source: STR achieves the highest throughput, while EMLSR stays close to SLO at low load and exceeds it under heavy OBSS load.
[31]	Fig. 4	4.7, 4.8	Consistent with the source: the A-MPDU ranking, the small gain from a 3 ms TXOP limit, and the decrease in throughput with increasing load are all reproduced.
[19]	Fig. 1, Fig. 2	4.9, 4.10	Reproduces the source: MLO STR delay remains well below SLO delay and grows slowly, while SLO delay increases sharply with load and with the number of stations.
[18]	Fig. 2	4.11	Matches the source: in contention-free conditions, additional links carry more throughput while maintaining similar p50 and p99 delays.
[17]	Fig. 4	4.12	Agrees with the source: STR significantly reduces delay, whereas EMLSR is comparable to, or slightly worse than, SLO.
[18]	Fig. 4	4.14	Reproduces the anomaly reported in the source: the 99th-percentile delay increases with the number of shared links, while the median delay improves only slightly.
[18]	Fig. 5	4.16	Reproduces the ordering observed in the source: STR:1+1 gives lowest tail delay, whereas fully shared STR:5 case gives highest tail delay.
[20]	Fig. 9	4.18a, 4.19c	Partially matches the source: the trend that MLO STR remains below SLO is observed at every occupancy level, but the asymmetric-occupancy anomaly reported in the reference is not reproduced.
[35]	Fig. 4	4.20a, 4.20b	Consistent with the source: MLO satisfies VR delay bounds at a lower MCS and channel width than SLO, with the uplink remaining the limiting direction.
[32]	Fig. 3–5	4.21–4.23	Differs from the source: the qualitative starvation of legacy stations agrees, but STR aggregate throughput obtained here increases and then decreases near 90% legacy traffic, unlike the monotonic decline reported in the source.

5. Summary

This thesis investigated the performance of multi-link operation in IEEE 802.11be networks through simulation in ns-3. A set of scenarios reproducing studies from recent literature was implemented and evaluated, comparing the two MLO modes supported by the simulator, STR and EMLSR, against single-link operation in terms of throughput, latency, and suitability for latency-critical traffic.

Firstly, the evaluation of the EMLSR parameters identified which of them actually matter: keeping the auxiliary radio able to follow the main one on a link switch was by far the most influential setting, while several other parameters had no measurable steady-state effect. These discoveries were crucial for the rest of the studies as they shaped the baseline parameters for EMLSR.

The throughput results confirmed the headline promise of MLO. The STR mode improves on SLO in every tested configuration, roughly in proportion to the number of links and the channel width, and most strongly in contention-free conditions. The EMLSR mode, which can use only one link at a time, behaved quite differently: close to SLO when the medium was free, and beneficial only when contention made the ability to pick a less busy link worthwhile, occasionally even falling slightly behind SLO.

The latency results drew a sharper line between the two operation types. In contention-free conditions the additional links act as spare capacity and MLO reduces the channel access delay substantially. Under contention the picture is more nuanced: the extra links become additional competitors, and the delay may grow rather than shrink unless the links are assigned with a specific, beneficial pattern. The configuration in which each network keeps one private link in addition to a shared one was the most effective at reducing the latency, whereas pooling all of the links among all the contending networks was the most inefficient way. These observations show that the way the links are assigned matters more than their quantity. In the channel occupancy scenario the second link proved beneficial in every tested combination, including the strongly asymmetric ones. The latency anomaly reported in the literature for asymmetrically occupied links was not reproduced by ns-3, suggesting that there may be some discrepancies between link assignment and contention handling between simulators.

The remaining scenarios addressed more specific questions. For virtual-reality traffic, MLO was shown to relax the physical layer settings required to meet the delay bounds, with the uplink emerging as the limiting direction. The coexistence study showed that the throughput gains of MLO come at a cost to the surrounding legacy stations, which are increasingly starved as the multi-link devices occupy several channels at once, and that fairness remains high only among devices of the same kind.

Taken together, the results largely confirm the trends reported in the literature: the throughput gains of STR, the conditional latency benefit of MLO, the importance of channel assignment under contention, and the impact of MLO on legacy coexistence, while also highlighting points where an independent simulator disagrees with the original studies, such as the absence of the asymmetric-occupancy latency anomaly. The results support the broader conclusion that MLO is a powerful but conditional mechanism, whose benefit depends strongly on the operating mode, on the load, and on the management of the links.

Several limitations of this work suggest directions for further study. The scenarios assume static stations and fixed-rate transmissions, so the interaction of MLO with mobility and with rate adaptation remains open. The traffic is generated mostly at the UDP level; the behaviour of MLO under TCP, whose acknowledgements are sensitive to how the links are used, would be the natural extension. Likewise, only the STR and EMLSR modes are available in the simulator, leaving NSTR, MLSR and EMLMR, as well as the many proposed traffic-to-link allocation and scheduling policies, for future evaluation. Extending the scenarios to larger topologies would further test the conclusions drawn here.

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